



CausalCACTUS: A Causally Consistent and Context-Aware Framework for Counterfactual Explanations

Zhendong Wang¹ · Diego García² · José M. Enguita² · Saikat Chatterjee¹ · Magnus Jansson¹

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Abstract

Counterfactual explanations have been proposed to provide actionable insights for complex black-box classifiers by suggesting “what-if” explanations that alter undesirable prediction outcomes to desired ones. Recent advances have emphasized desirable properties such as causality, plausibility, and contextual alignment; however, existing methods often address these objectives in isolation. In particular, causal counterfactual approaches enforce structural dependencies among features, typically assuming access to a fully specified structural causal model, while context-aware counterfactual frameworks focus on improving alignment with user-defined contextual constraints. As a result, a gap remains between context-aware and causally grounded counterfactual generation, especially in latent-space representations, which can jointly address multiple objectives and improve validity across domains. In this work, we introduce CausalCACTUS, a causally consistent and context-aware counterfactual explanation framework that extends the CACTUS method by integrating automatically learned causal structure into latent-space counterfactual generation using a composite β -VAE model. CausalCACTUS incorporates a directed acyclic graph learned from data-driven causal discovery to impose causal feasibility constraints, together with a phase-based latent optimization strategy that improves convergence stability and counterfactual validity while preserving user-defined contextual constraints. Our experimental evaluation across five publicly available tabular datasets and multiple baseline models demonstrates that CausalCACTUS achieves improved context alignment and causal consistency metrics while maintaining competitive performance on standard counterfactual metrics, such as validity and proximity. Ablation studies and qualitative examples further illustrate that jointly enforcing context preservation and causal structure leads to more actionable, trustworthy counterfactual explanations that respect user-defined contexts and causal dependencies in real-world scenarios.

Keywords Counterfactual explanations · Structural causal models · Context-aware systems · Deep learning · Model interpretability

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Extended author information available on the last page of the article

1 Introduction

Deep learning models have demonstrated competitive performance across a wide range of prediction domains, including tasks like image, time series, and tabular classification (Borisov et al., 2024; Chai et al., 2021; Ismail Fawaz et al., 2019). However, these models are often considered “black boxes”, which compromises model interpretability and makes it difficult to understand model predictions. To mitigate this issue, previous work in eXplainable AI (XAI) has been introduced, including feature importance methods like Shapley values (SHAP) and saliency maps, as well as counterfactual explanations (Guidotti, 2022; Lundberg & Lee, 2017; Molnar, 2025). Among these, counterfactual explanations have gained particular attention because they provide actionable, instance-level explanations that align with “right to explanation” regulations, such as the General Data Protection Regulation (GDPR) (Wachter et al., 2017). Counterfactuals offer post-hoc explanations for already deployed black-box models without revealing their internal structures, while providing actionable “what-if” insights into the model’s reasoning process (Guidotti, 2022; Verma et al., 2024). In practical settings such as loan applications, counterfactuals can suggest how applicants with a negative outcome (i.e., loan rejection) could make actionable changes to their profiles to obtain a favorable decision (i.e., loan approval).

Recent counterfactual research has emphasized incorporating additional desiderata beyond validity and proximity to guide the generation of counterfactual instances, including *causality* and *actionability* (Verma et al., 2024). Causal counterfactual methods aim to prevent unrealistic recommendations, such as immutable feature changes (e.g., requesting the applicant to become younger), by enforcing causal constraints through structural causal models (SCMs) or explicit structural interventions (Duong et al., 2024; Karimi et al., 2021; Mahajan et al., 2019; Nogueira et al., 2022). While these approaches improve feasibility and actionability, they typically assume access to a known or fully specified SCM, which is often unavailable in real-world applications. Moreover, most causal counterfactual methods optimize directly in the feature space, limiting their applicability to handle complex data representations and latent-space optimization. To mitigate these challenges, recent research on *context-aware* learning integrates context information (e.g., user-related aspects and socioeconomic situations) through explicit modeling to enhance machine learning tasks for accurate scenario interpretation (Jiang et al., 2022; Messmer et al., 2024). At the same time, previous works investigated the generation of *plausible* counterfactuals that adhere to the underlying data manifold, including nearest-neighbor estimators (Kanamori et al., 2020; Laugel et al., 2019), density estimation measures (Brugmans et al., 2024; Kenny & Keane, 2021; Pawelczyk et al., 2020), and class prototypes (Van Looveren & Klaise, 2021).

Additionally, earlier studies explored leveraging deep generative models to guide counterfactual search through learned latent representations, including auto-encoders (AEs) and variational auto-encoders (VAEs) (Pawelczyk et al., 2020; Van Looveren & Klaise, 2021; Wang et al., 2024, 2021). Latent-space approaches demonstrated promising performance because they encouraged proximate and realistic changes following high-density regions and could generalize across modalities, including time series and images (Guidotti, 2022). However, optimizing plausibility and proximity in the feature space alone does not guarantee alignment with user-defined contexts, such as demographic or socioeconomic conditions; nor does it ensure causal dependencies among features. This ongoing challenge motivates the need for counterfactual explanation frameworks that jointly optimize for con-

text preservation and causal consistency using a latent representation (Verma et al., 2024), which is the central focus of this work.

Towards the direction of generating feasible counterfactuals, the CACTUS framework (Pérez et al., 2025) introduced context-aware counterfactual generation by providing explicit control over user-defined context information. By adapting a context-conditional latent representation β -VAE model, their approach supports both context-preserving and context-changing scenarios tailored to user-specific preferences. Their evaluations on tabular and image classification tasks demonstrated competitive performance in context validity compared to existing counterfactual methods. However, CACTUS does not explicitly model causal dependencies among features. As a result, generated counterfactuals may violate structural relationships; in other words, the method may modify causally dependent variables without corresponding changes in their ancestors. In contrast, existing causality-related counterfactual methods rely on a fully specified SCM to generate counterfactual explanations (Duong et al., 2024; Karimi et al., 2021; Mahajan et al., 2019), such that there remains a gap between the context-aware objective and causally grounded counterfactual generation. This limitation is particularly critical in high-stakes domains such as credit loan applications, where causal feature dependencies must be respected to provide trustworthy explanations (Verma et al., 2024). To our knowledge, there has not been any existing work that is capable of generating counterfactuals that satisfy both objectives in this setting.

To address this gap, we propose CausalCACTUS, a *Causally Consistent and Context-aware Framework* by extending CACTUS (Pérez et al., 2025), where we integrate automatically learned causal structure into latent-space counterfactual generation. CausalCACTUS augments the original framework with three key components: (1) incorporating a data-driven causal discovery approach to learn a directed acyclic graph (DAG), (2) imposing causal feasibility constraints derived from the DAG to encourage structural feature changes, and (3) adopting a phase-based optimization strategy that improves the convergence stability and validity of generated counterfactuals. Importantly, our approach does not require a fully specified SCM; instead, it can incorporate causal graphs obtained using state-of-the-art causal discovery methods such as ReX (Renero et al., 2026).

Figure 1 presents an illustrative example of the proposed CausalCACTUS framework in a credit decision scenario. Given an original instance predicted as “*loan risk*” with a user-defined context (Age>50), the CausalCACTUS explainer generates a counterfactual that modifies actionable features leading to the prediction label “*no loan risk*”, while respecting the user-defined context and given causal feature dependency between monthly income and the number of credit lines. The blue-colored feature changes highlight how CausalCACTUS generates counterfactuals that fulfill prediction validity, context preservation, and causal consistency simultaneously. To further highlight improved causal consistency and context alignment, our qualitative analyses provide counterfactual examples from our experiments for an empirical evaluation.

1.1 Our contributions

We summarize our contributions as follows:

1. We introduce CausalCACTUS, a causally consistent and context-aware framework for counterfactual explanations that searches in a latent representation and jointly optimizes

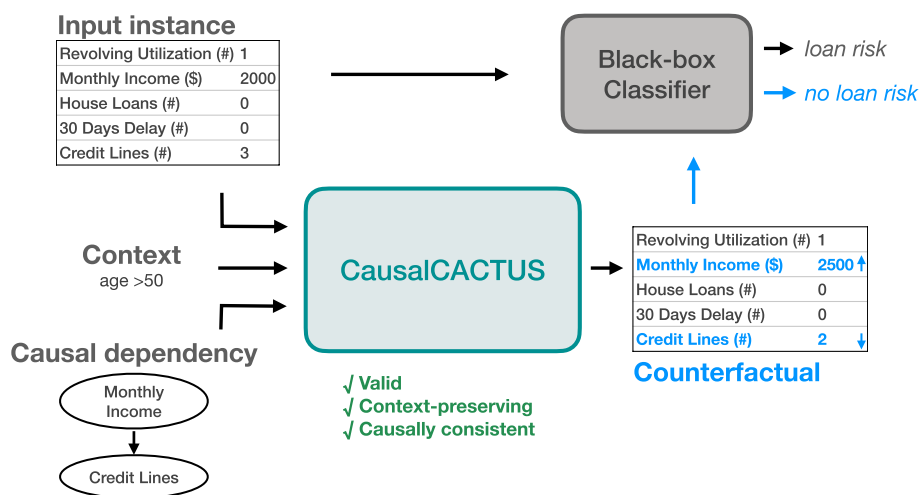


Fig. 1 An illustrative example of the CausalCACTUS counterfactual generation for credit decision classification. Starting from an original input instance classified as “loan risk”, the counterfactual explanation modifies actionable features to achieve the target prediction as “no loan risk”. The highlighted feature changes (blue-colored) of the generated counterfactual respect the provided causal dependency (i.e., between monthly income and credit lines) while preserving the user-defined context (age >50)

objectives for user-defined context alignment and structural constraints of causal feature dependencies.

2. We extend the original CACTUS framework by explicitly incorporating data-driven causal dependencies as feasibility constraints into latent-space counterfactual generation. Specifically, we incorporate a DAG learned by the novel causal discovery ReX method, without requiring a fully specified SCM. We also introduce two augmented components: causal feasibility constraints and a phase-based latent optimization strategy, both of which improve optimization convergence stability and counterfactual validity.
3. We conduct a comprehensive evaluation on five publicly available tabular datasets across diverse application domains against four baseline models, demonstrating that CausalCACTUS achieves competitive metrics for compactness, proximity, contextual alignment, and causal consistency, while maintaining strong performance on counterfactual validity.
4. We provide ablation studies examining the effects of the α loss parameter, as well as the individual and combined components of causal constraints and phase-based optimization in CausalCACTUS, highlighting favorable trade-offs among validity, context alignment, and causal consistency. We additionally analyze qualitative examples that show how explicitly integrating causal dependencies and context-preserving optimization leads to counterfactual explanations that better align with real-world feasibility constraints.

2 Related Work

Prior work on causal counterfactual explanations has primarily focused on imposing *feasibility* and *actionability* by incorporating causal reasoning into the counterfactual generation process, often through SCMs. Mahajan et al. (2019) proposed incorporating predefined SCMs into feature-space optimization via a causal proximity regularizer, which preserves immutable causal constraints (e.g., age and sex groups) while modifying only mutable features. Karimi et al. (2021) highlighted that generating counterfactuals via explicit structural interventions through a known SCM, such that the recommended actions in algorithmic recourse are causally valid and cost-minimal. ProCE (Duong et al., 2024) was proposed as a multi-objective framework using evolutionary optimization (NSGA-II) that jointly incorporates a prototype loss and a causality-preserving loss derived from a known SCM. However, a key limitation of these works is the assumption that the SCM and structural equations are known or explicitly specified, which may not hold in many real-world applications. To mitigate this issue, Krauss et al. (2025) introduced CausalCSSE, which uses LiNGAM (Nogueira et al., 2022) to discover the causal structure and apply a genetic algorithm to search for causal counterfactuals. The discovered graph is applied as a hard constraint during candidate generation and evaluation, focusing on causal validity in feature space. This work highlights the potential of integrating novel causal discovery methods, such as ReX (Renero et al., 2026) and variants of LiNGAM (Nogueira et al., 2022), into counterfactual generation to improve causal consistency and interpretability, without requiring full access to an explicitly provided SCM. Furthermore, existing approaches remain focused on causal structural validity, offering limited mechanisms to explicitly control context-aware and user-specific feasibility.

Context-aware adaptation is crucial in machine learning applications, especially when context information can accurately interpret domain scenarios (Jiang et al., 2022). Such approaches integrate context information (e.g., user-related aspects) through various modalities to, for instance, enhance recommendation systems (Messmer et al., 2024) or select relevant features for classification tasks (Wang & Bilgic, 2023). Previous researchers proposed context-aware adaptation in post-hoc XAI methods. For instance, context-aware learning can capture feature dependencies for local interpretable model-agnostic explanations (LIME) (Anwar et al., 2023), and context-aware SHAP can complement feature selection, thereby enhancing model fairness and robustness (Vellaichamy et al., 2024). More recently, CACTUS (Pérez et al., 2025) was proposed to incorporate context-awareness via multi-objective gradient-based optimization in latent space for counterfactual generation. Their approach provides more feasible counterfactuals aligned with user preferences, but at the cost of validity as a trade-off. To our knowledge, there is a gap in mechanisms that effectively incorporate both causal consistency and context-awareness to support user-centric feasibility while respecting causal structures during counterfactual generation.

Another group of counterfactual methods explored *plausibility* to ensure that counterfactuals closely resemble the ground-truth data manifold. Laugel et al. (2019) introduced plausibility by employing local outlier factor (LOF) as a distance measure in a distribution-aware approach; the DACE method incorporated LOF through mixed-integer linear optimization, with both methods ensuring counterfactuals aligned with the target class distribution (Kanamori et al., 2020). Recent research has explored latent representations with counterfactual plausibility in various prediction domains. Pawelczyk et al. (2020) adapted

plausibility and proposed connectedness in the C-CHVAE model to assess the proportion of counterfactuals using correctly classified instances for tabular credit prediction, employing a VAE model and a Gaussian mixture prior to prevent immutable feature changes. The PrototypeCF method defined a reconstruction error term from a learned AE and class prototypes to prevent out-of-distribution counterfactual generation, guiding the search for more plausible counterfactuals for image and tabular classification (Van Looveren & Klaise, 2021). The NICE model incorporated an AE with nearest-neighbors in optimization to generate plausible counterfactual instances, demonstrating a trade-off between plausibility and proximity (Brugmans et al., 2024). LatentCF++ proposed instantiations using convolutional neural networks (CNN) and long short-term memory (LSTM) AEs to compare the performance trade-offs for time series classification (Wang et al., 2021); this was further generalized as the unconstrained version of the Glacier method (Wang et al., 2024). Both PrototypeCF and LatentCF++ serve as suitable baselines compared to CausalCACTUS and the original CACTUS in our empirical evaluation.

3 Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^n$ be an n -dimensional feature vector comprising n features as $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$. The feature vector $\mathbf{x} \sim P(\mathbf{x})$ and $\mathbf{x}^{(i)}$ denotes i -th instance of the vector $\mathbf{x}$. Let $\mathcal{D} = \{\mathbf{x}^{(i)}\}_{i=1}^N$, where $\mathbf{x}^{(i)} \in \mathbb{R}^n$, denote a dataset of N instances, and the train-test split is defined as $\mathcal{D}_{\text{train}} = \{\mathbf{x}^{(i)}\}_{i=1}^{N_{\text{train}}}$ and $\mathcal{D}_{\text{test}} = \{\mathbf{x}^{(i)}\}_{i=N_{\text{train}}+1}^N$. Let $y \in \{0, 1\}$ denote the ground-truth label, where $y = 1$ indicates that $\mathbf{x}$ belongs to the positive class and $y = 0$ to the negative class. We define $f: \mathbb{R}^n \rightarrow [0, 1]$ as a trained binary classifier that maps an input instance $\mathbf{x}$ to the predicted probability $\hat{y}$, such that $\hat{y} = f(\mathbf{x})$. The predicted class label is defined as $\hat{y}_{\text{la}} = \mathcal{I}(\hat{y} \geq \tau)$, where $\tau \in [0, 1]$ is a fixed decision threshold and $\mathcal{I}(\cdot)$ denotes the indicator function. To model causal feature dependencies, we assume prior knowledge of a DAG $\mathcal{G} = (V, \mathcal{E})$, where V is a subset of the n features from $\mathbf{x}$ and $\mathcal{E}$ is the set of direct edges, where $V \subseteq \{x_1, x_2, \dots, x_n\}$ and $\mathcal{E} \subseteq V \times V$. For each feature x_j , its parent set is defined as $\text{Pa}(x_j) = \{x_h \in V : (x_h \rightarrow x_j) \in \mathcal{E}\}$, where a direct edge $x_h \rightarrow x_j$ indicates that “feature x_h causes feature x_j ”. Consistent with causal discovery settings, we assume access to graph structure (i.e., parent–child relations) from a DAG $\mathcal{G}$, and we do not assume access to structural equations or a fully specified SCM. Additionally, each instance $\mathbf{x}$ is associated with a multi-label context vector $\mathbf{c} = [c_1, \dots, c_d]$, where d denotes the total number of contextual conditions. Each component $c_i \in \{0, 1\}$ indicates whether the input instance belongs to the i -th contextual condition. Specifically, an input instance $\mathbf{x}$ may be assigned to one or more contexts; for example, a loan applicant might simultaneously belong to two different binary contexts: $\text{age} > 50$ ($c_1 = 1$) and $\text{number of dependents} \geq 1$ ($c_2 = 1$).

In this paper, we formulate the problem of generating context-preserving and causally consistent counterfactuals for binary classification models. Our objective is for the counterfactual generation to incorporate user-defined contextual information and respect known causal dependencies among features. Figure 2 provides an overview of the proposed problem setting. The figure illustrates a counterfactual $\mathbf{x}_{\text{cf}}$ that changes the classifier prediction while remaining within the original context region and satisfying the causal feature dependencies. For example, in the credit decision prediction scenario, a desirable counterfactual should cross the decision boundary within the context region of $\text{Age} > 50$ ($c_1 = 1$), while

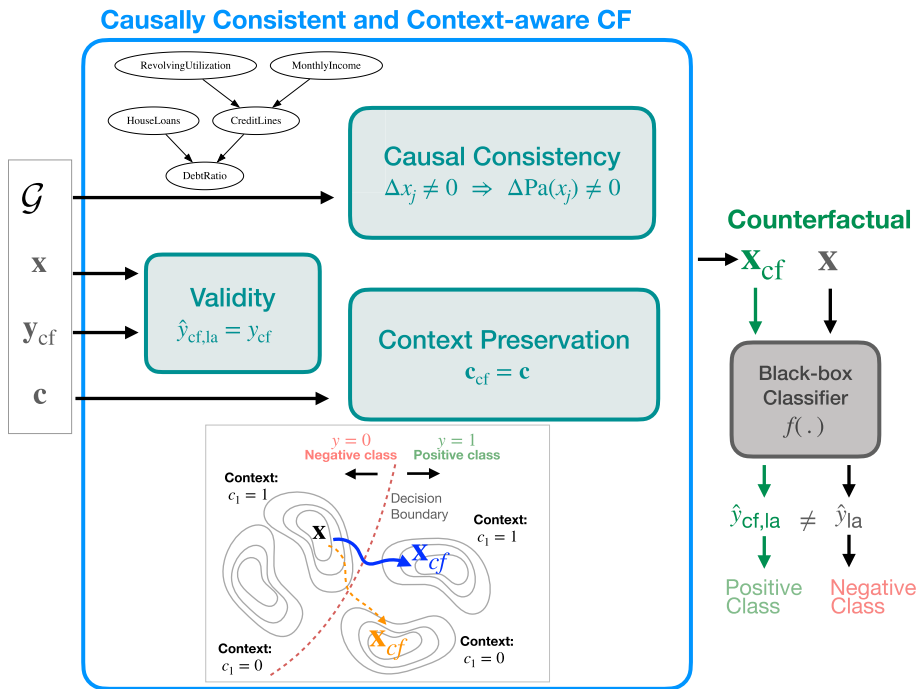


Fig. 2 Illustration of the causally consistent and context-preserving counterfactual generation problem. Given an input instance x , its context vector c , a target label y_{cf} , and a DAG $\mathcal{G}$, the objective is to generate x_{cf} that satisfies three properties: validity, causal consistency, and context preservation, as shown in the green blocks. The lower panel provides a geometric interpretation for context preservation: the blue trajectory crosses the decision boundary while preserving the context ($c_1 = 1$), whereas the dashed orange trajectory represents a context-changing alternative

adhering to the causal dependency structure encoded by the DAG $\mathcal{G}$. This objective is formally defined in two problem statements.

Problem 1 *Context-preserving counterfactual explanations.* Given $f(\cdot)$, an input instance x , and its context label c such that $\hat{y} = f(x)$ with corresponding label $\hat{y}_{la}$, we define a context-preserving counterfactual instance x_{cf} as:

$$\hat{y}_{cf} = f(x_{cf}) \text{ s. t. } \hat{y}_{cf,la} \neq \hat{y}_{la}, \text{ dist}(x, x_{cf}) \text{ is minimized, and } c_{cf} = c,$$

where the predicted label $\hat{y}_{cf,la} = \mathcal{I}(\hat{y}_{cf} \geq \tau)$ of instance x_{cf} leads to a different prediction (validity), the distance constraint $\text{dist}(\cdot, \cdot)$ encourages x_{cf} to be as close as possible to the original instance (proximity), and c_{cf} is the context condition associated with x_{cf} . We refer to the constraint $c_{cf} = c$ as context validity, indicating that the counterfactual must remain within the user-specified original context.

Problem 2 *Causally consistent and context-preserving counterfactual explanations.* We additionally require the counterfactual to respect the known feature dependency structure from a provided DAG $\mathcal{G}$. For any feature x_j with parent set $\text{Pa}(x_j)$, we define the feature change as: $\Delta x_j = |x_{cf,j} - x_j|$. We define a counterfactual as causally consistent to $\mathcal{G}$ if:

$$\Delta x_j \neq 0 \Rightarrow \exists x_h \in \text{Pa}(x_j) \text{ s.t. } \Delta x_h \neq 0.$$

That is, the change of x_j may happen only if at least one of its parent features x_h also changes with respect to $\mathcal{G}$. Therefore, we define a context-preserving and causally consistent counterfactual $\mathbf{x}_{\text{cf}}$ as:

$$\begin{aligned} \hat{y}_{\text{cf}} = f(\mathbf{x}_{\text{cf}}) \text{ s.t. } \hat{y}_{\text{cf},\text{la}} \neq \hat{y}_{\text{la}}, \text{ dist}(\mathbf{x}, \mathbf{x}_{\text{cf}}) \text{ is minimized,} \\ \mathbf{c}_{\text{cf}} = \mathbf{c}, \text{ and } \mathbf{x}_{\text{cf}} \text{ is causally consistent to } \mathcal{G}. \end{aligned}$$

The formulation could be extended to allow $\mathbf{c}_{\text{cf}} \neq \mathbf{c}$, i.e., the context-changing scenario defined by Pérez et al. (2025); however, in this work, we focus on context-preserving and causally consistent counterfactual explanations, which are more relevant for actionable decision support and user-specified constraints.

4 Causally Consistent and Context-Aware Counterfactual Explanation Framework

We propose *Causally Consistent and Context-aware Counterfactual Explanations* (Causal-CACTUS), a framework for generating context-aware and causally consistent counterfactual explanations that extends the previously introduced CACTUS framework (Pérez et al., 2025). In this section, we first introduce the original CACTUS framework, illustrate the framework in detail, outline its main components with deep learning instantiations; and then describe how it is extended to CausalCACTUS with data-driven causal discovery and augmented modeling components.

Input: Original sample $\mathbf{x}_0$, target label y_{cf} , target context $\mathbf{c}_{\text{cf}}$, encoder E , decoder D , classifier f , context classifier f_{ctx} , learning rate η , loss weights α, γ , epochs T

Output: Counterfactual $\mathbf{x}_{\text{cf}}$

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1:  $\mathbf{z}_0 \leftarrow E(\mathbf{x}_0)$ 
2:  $L_{\text{ctx}}^{\text{best}}, L_{\text{dist}}^{\text{best}}, \delta_y^{\text{best}} \leftarrow +\infty$ 
3: for  $k = 1$  to  $T$  do
4:    $L_{\text{total}} \leftarrow \mathcal{L}_{\text{total}}(\mathbf{x}_0, \mathbf{x}_{k-1}, \mathbf{z}_{k-1}, \hat{y}_{k-1}, \hat{\mathbf{c}}_{k-1})$ 
5:    $\mathbf{z}_k \leftarrow \mathbf{z}_{k-1} - \eta \cdot \frac{\nabla_{\mathbf{z}} L_{\text{total}}}{\max(|\nabla_{\mathbf{z}} L_{\text{total}}|)}$ 
6:    $\mathbf{x}_k \leftarrow D(\mathbf{z}_k)$ 
7:    $\hat{y}_k \leftarrow f(\mathbf{x}_k), \quad \hat{\mathbf{c}}_k \leftarrow f_{\text{ctx}}(\mathbf{z}_k)$ 
8:    $L_{\text{ctx}} \leftarrow \mathcal{L}_{\text{ctx}}(\hat{\mathbf{c}}_k, \mathbf{c}_{\text{cf}})$ 
9:    $L_{\text{dist}} \leftarrow \mathcal{L}_{\text{dist}}(\mathbf{x}_0, \mathbf{x}_k)$ 
10:  if  $|y_{\text{cf}} - \hat{y}_k| \leq \delta_y^{\text{best}}$  and  $L_{\text{ctx}} \leq L_{\text{ctx}}^{\text{best}}$  then
11:     $\mathbf{x}_{\text{cf}} \leftarrow \mathbf{x}_k$ 
12:     $\mathbf{z}_{\text{cf}} \leftarrow \mathbf{z}_k$ 
13:     $L_{\text{ctx}}^{\text{best}} \leftarrow L_{\text{ctx}}$ 
14:     $L_{\text{dist}}^{\text{best}} \leftarrow L_{\text{dist}}$ 
15:     $\delta_y^{\text{best}} \leftarrow |y_{\text{cf}} - \hat{y}_k|$ 
16:  end if
17: end for

```

Algorithm 1 CACTUS counterfactual generation

4.1 CACTUS: Context-Aware Counterfactuals

4.1.1 Counterfactual Search in Latent Space

Context-Aware CounTerfactUals (CACTUS) (Pérez et al., 2025) is proposed to ensure plausible counterfactuals and to avoid unrealistic outputs by generating them within a latent space learned by an AE-based model. As illustrated in Fig. 3a, an original input instance x_0 is mapped into the latent space by a pre-trained encoder $E(\cdot)$ via $\underline{z} = E(\underline{x})$, where $\underline{z} \in \mathbb{R}^m$ denotes the representation of the original instance in a compact m -dimensional latent space, i.e., $m \ll n$. Ideally, $\underline{z}$ captures the most relevant and underlying semantical information of the input, forming a manifold representation of the input space. Hence, performing the

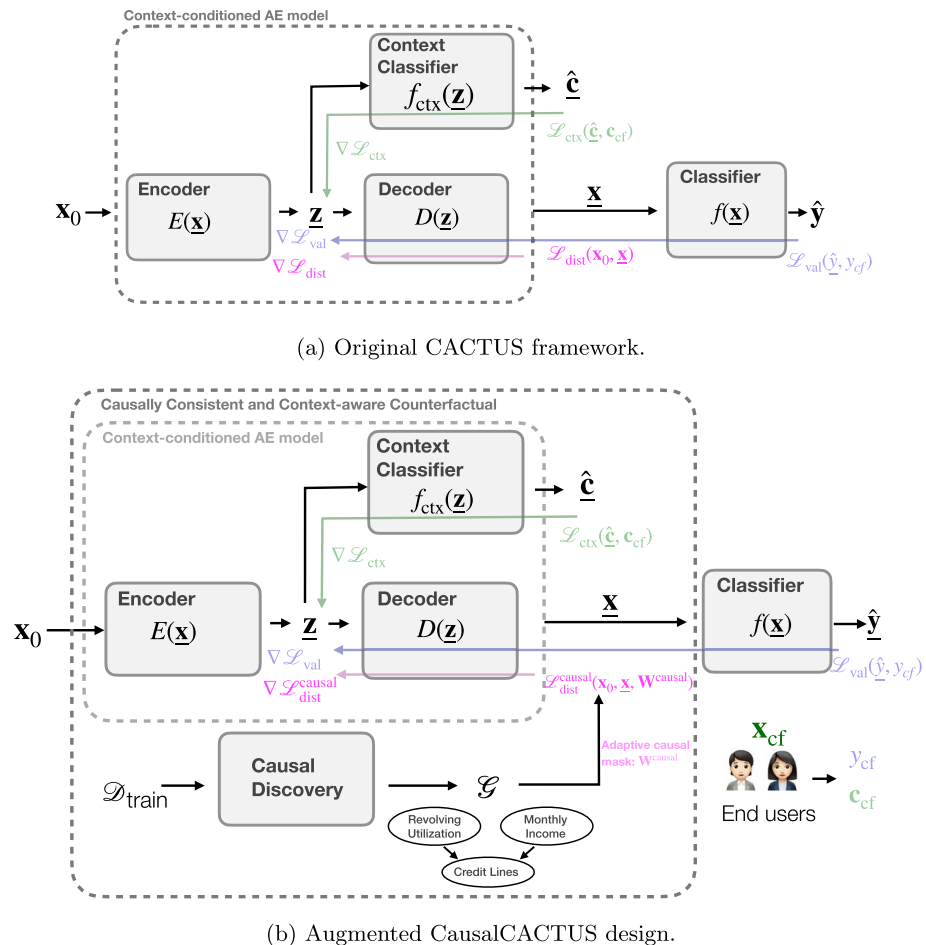


Fig. 3 Overview of the original CACTUS framework (a) and the augmented CausalCACTUS design (b). In the original CACTUS workflow, a context-conditional AE-based model guides the latent representation $\underline{z}$ by a context classification subtask to integrate user-defined context preferences into counterfactual generation. CausalCACTUS extends this pipeline by incorporating a learned causal graph $\mathcal{G}$ via a causal discovery component and an additional causal distance loss, ensuring that feature changes follow learned dependencies while still achieving the target label and context

counterfactual search in this manifold leads to more realistic modifications to the input. Accordingly, we utilize an instantiation of CACTUS using a β -VAE model (Burgess et al., 2018) as the AE-based architecture, due to its enhanced disentanglement capabilities, which improve counterfactual generation. Nevertheless, the CACTUS framework is agnostic to any latent-space model and can be instantiated with alternative AE variants.

Algorithm 1 illustrates the counterfactual search procedures: starting from an initial latent point $\mathbf{z}_0 = E(\mathbf{x}_0)$, the algorithm iteratively explores candidate solutions using the decoder, i.e., $\mathbf{x} = D(\mathbf{z})$. Specifically, at the k -th iteration, the decoder output is denoted as $\mathbf{x}_k = D(\mathbf{z}_k)$, which represents a counterfactual candidate $\mathbf{x}$ iteratively moving toward satisfying the desired counterfactual properties. We propose a multi-objective loss function that evaluates how a counterfactual candidate satisfies the validity, proximity, and context validity requirements, defined as follows:

Validity loss. The validity loss $\mathcal{L}_{\text{val}}$ measures the difference between the classifier prediction $\hat{y} = f(\mathbf{x})$ and the target label y_{cf} , where the candidate $\mathbf{x}$ is obtained by mapping $\mathbf{z}$ back to the input space using $\mathbf{x} = D(\mathbf{z})$. To this end, we adopt the *binary cross-entropy* loss function:

$$\mathcal{L}_{\text{val}}(\hat{y}, y_{\text{cf}}) = -y_{\text{cf}} \log(\hat{y}) - (1 - y_{\text{cf}}) \log(1 - \hat{y}), \quad \hat{y} = f(\mathbf{x}). \quad (1)$$

This term penalizes candidate instances that do not lead the classifier prediction $\hat{y}$ to the desired counterfactual label.

Distance loss. We incorporate a *distance loss* term that penalizes extensive modifications and encourages proximity. Accordingly, we utilize the Euclidean distance between the original input $\mathbf{x}_0$ and the counterfactual candidate $\mathbf{x}$ as a proximity constraint:

$$\mathcal{L}_{\text{dist}}(\mathbf{x}_0, \mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|_2, \quad \mathbf{x} = D(\mathbf{z}). \quad (2)$$

Context loss. To enforce context validity, the proposed CACTUS generates counterfactuals within a latent space that is explicitly structured to encode contextual information. To this end, we utilize a modification to the standard AE architecture by adding an auxiliary classification head that estimates the context label vector $\hat{\mathbf{c}}$ of the input $\mathbf{x}$ from its latent representation $\mathbf{z}$, i.e., $\hat{\mathbf{c}} = f_{\text{ctx}}(\mathbf{z})$, where the context classifier $f_{\text{ctx}}(\cdot)$ estimates the context label using a standard neural network composed of fully-connected layers. The overall architecture, including the auxiliary context classifier, is illustrated in Fig. 3a. This composite architecture is jointly trained prior to counterfactual generation.

The estimated context vector $\hat{\mathbf{c}}$ during counterfactual search can be compared to the target context $\mathbf{c}_{\text{cf}}$ to evaluate whether it satisfies the user-specified context. As a result, we introduce a context loss function $\mathcal{L}_{\text{ctx}}$ based on the binary cross-entropy between $\hat{\mathbf{c}}$ and $\mathbf{c}_{\text{cf}}$:

$$\mathcal{L}_{\text{ctx}}(\hat{\mathbf{c}}, \mathbf{c}_{\text{cf}}) = -\mathbf{c}_{\text{cf}} \log(\hat{\mathbf{c}}) - (1 - \mathbf{c}_{\text{cf}}) \log(1 - \hat{\mathbf{c}}), \quad \hat{\mathbf{c}} = f_{\text{ctx}}(\mathbf{z}). \quad (3)$$

This loss guides the optimization toward counterfactuals that satisfy the context validity requirement in the context-preserving ($\mathbf{c}_{\text{cf}} = \mathbf{c}_0$) scenario.

Finally, the validity, distance, and context loss terms are combined into a unified multi-objective loss function¹:

$$\mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\hat{\mathbf{y}}}, \underline{\hat{\mathbf{c}}}) = \alpha \mathcal{L}_{\text{val}}(\underline{\hat{\mathbf{y}}}, y_{\text{cf}}) + (1 - \alpha) \mathcal{L}_{\text{ctx}}(\underline{\hat{\mathbf{c}}}, c_{\text{cf}}) + \gamma \mathcal{L}_{\text{dist}}(x_0, \underline{\mathbf{x}}), \quad (4)$$

where the user-defined hyperparameters α and γ provide the flexibility to control the relative importance of the validity and context loss terms, and the distance loss, respectively. We conduct an ablation study to illustrate the hyperparameter effects of the proposed framework in our experimental evaluation.

4.1.2 Gradient-Based Search

The search in the latent space is conducted by minimizing the loss term in Eq. 4, which jointly encodes predictive, contextual, and proximity objectives:

$$z_{\text{cf}} = \arg \min_z \mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\hat{\mathbf{y}}}, \underline{\hat{\mathbf{c}}}), \quad (5)$$

which produces $x_{\text{cf}} = D(z_{\text{cf}})$, $\hat{y}_{\text{cf}} = f(x_{\text{cf}})$ and $\hat{c}_{\text{cf}} = f_{\text{ctx}}(z_{\text{cf}})$, respectively. Following the pseudo-code in Algorithm 1, the initial point z_0 is iteratively updated according to the *gradient descent* rule in the minimization problem. Specifically, at k -th iteration step, the update step is represented as:

$$z_k = z_{k-1} - \eta \frac{\nabla \mathcal{L}_{\text{total}}(x_0, x_{k-1}, z_{k-1}, \hat{y}_{k-1}, \hat{c}_{k-1})}{\max(|\nabla \mathcal{L}_{\text{total}}(x_0, x_{k-1}, z_{k-1}, \hat{y}_{k-1}, \hat{c}_{k-1})|)}, \quad (6)$$

where η is the learning rate. To ensure stable updates during optimization, the gradient direction is normalized at each step so that updates follow the direction of descent while maintaining a fixed maximum step size, independent of the gradient's magnitude. The update rule in Eq. 6 is executed for a total of T maximum iterations. At the k -th step, the candidate x_k , obtained from z_k via the decoder, is stored if the corresponding loss scores improve upon the previous best in terms of prediction and context validity.

4.2 CausalCACTUS: Causally Consistent and Context-Aware Counterfactuals

We propose CausalCACTUS, extending the original CACTUS (Pérez et al., 2025) and augmented with a data-driven causal discovery approach and feasibility constraints to guide the counterfactual generation. Figure 3 illustrates the difference between the original CACTUS pipeline and the augmented CausalCACTUS. The benefit of CausalCACTUS is to prevent causally inconsistent counterfactuals while preserving context-conditioned latent optimization in the original framework. Next, we first describe the adapted causal discovery method using ReX and then introduce two augmented components: causal feasibility constraints and phase-based optimization.

¹ The expression $\mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\hat{\mathbf{y}}}, \underline{\hat{\mathbf{c}}}, y_{\text{cf}}, c_{\text{cf}}, \alpha, \gamma)$ is simplified as $\mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\hat{\mathbf{y}}}, \underline{\hat{\mathbf{c}}})$ throughout the manuscript to improve the readability of the notation.

Input: $\mathcal{D}_{\text{train}} = \{\mathbf{x}^{(i)}\}_{i=1}^{N_{\text{train}}}$, maximum iterations T_b , threshold ν

Output: DAG $\mathcal{G}$

```

1: Initialize  $A \in \mathbb{R}^{n \times n} \leftarrow 0$ 
2: for each  $x_j \in V$  do
3:   Train regressors  $\hat{f}_j$  using  $V \setminus \{x_j\}$ 
4: end for
5: for  $t = 1$  to  $T_b$  do
6:   Sample  $\mathcal{D}_{\text{train}}^{(t)} \subset \mathcal{D}_{\text{train}}$ 
7:   for each  $x_j \in V$  do
8:     Compute SHAP  $\Phi^{(j)} = \{\phi_h^{(j)} : h \neq j\}$ 
9:      $Pa_j \leftarrow \text{high-SHAP cluster}(\Phi^{(j)})$ 
10:    for  $x_h \in Pa_j$  do  $A_{h,j} \leftarrow A_{h,j} + 1$ 
11:    end for
12:   end for
13: end for
14:  $A \leftarrow A/T_b$ ; keep edges with  $A_{h,j} \geq \nu$ 
15:  $\mathcal{G} \leftarrow \text{union over regressors (DFN and GBT)}$ 
16: while cycle exists do remove edge with the largest SHAP discrepancy
17: end while

```

Algorithm 2 ReX causal discovery (Renero et al., 2026)

4.2.1 Causal Discovery Using ReX

The task of causal discovery is to construct a DAG $\mathcal{G} = (V, \mathcal{E})$ from the training data $\mathcal{D}_{\text{train}}$ to obtain the feature set V as a subset of n features, and $\mathcal{E}$ representing a set of directed edges over V . The ReX method (Renero et al., 2026) is a data-driven causal discovery algorithm that utilizes Shapley values from SHAP (Lundberg & Lee, 2017) using predictive models to identify conditional dependencies and extract the DAG $\mathcal{G}$. Algorithm 2 summarizes the approach of adapting ReX to generate a DAG into the CausalCACTUS framework. First, regressors $\hat{f}_j$ are trained for each feature x_j , using either a deep feedforward network (DFN) or gradient boosted trees (GBT). Next, bootstrapped SHAP values $\Phi^{(j)}$ are computed to measure the marginal contribution of x_h in predicting x_j . The cluster with the highest SHAP value is selected as a candidate parent, Pa_j , to update the matrix $A_{h,j}$. After reaching the maximum bootstrap iterations (denoted as T_b), only edges whose frequency exceeds the threshold ν are preserved. Finally, all candidate DAGs are combined via union into a final DAG $\mathcal{G}$. If a cycle (i.e., a loop of parent–child relationships) appears, the edge with the largest SHAP discrepancy score is removed until the DAG is acyclic, and $\mathcal{G}$ is returned.

With the adaptation of ReX, we directly incorporate learned causal relations from DAG $\mathcal{G}$ into downstream tasks in the CausalCACTUS framework. This does not require a fully specified SCM as a prerequisite. Additionally, ReX supports structural prior knowledge as constraints to guide the hierarchy of the learned DAG, and we demonstrate two tiers of priors as instantiations in our experiments.

Input: Original sample $\mathbf{x}_0$, target label y_{cf} , target context $\mathbf{c}_{\text{cf}}$, decision threshold τ , encoder E , decoder D , classifier f , context classifier f_{ctx} , learning rate η , loss weights α, γ , epochs T , feature bounds $[\mathbf{x}_{\min}, \mathbf{x}_{\max}]$, learned DAG $\mathcal{G}$, causal masking weight ϵ

Output: Counterfactual $\mathbf{x}_{\text{cf}}$

```

1:  $\mathbf{z}_0 \leftarrow E(\mathbf{x}_0)$ 
2:  $\mathbf{W}^{\text{causal}} \leftarrow \mathbf{1}$ 
3:  $L_{\text{total}}^{\text{best}}, L_{\text{ctx}}^{\text{best}} \leftarrow +\infty$ 
4: for  $k = 1$  to  $T$  do
5:    $\mathbf{x}_{k-1} \leftarrow D(\mathbf{z}_{k-1})$ 
6:    $\mathbf{x}_{k-1} \leftarrow \text{clip}(\mathbf{x}_{k-1}, [\mathbf{x}_{\min}, \mathbf{x}_{\max}])$ 
7:    $L_{\text{total}} \leftarrow \mathcal{L}_{\text{total}}(\mathbf{x}_0, \mathbf{x}_{k-1}, \mathbf{z}_{k-1}, \hat{y}_{k-1}, \hat{\mathbf{c}}_{k-1}, \mathbf{W}^{\text{causal}})$ 
8:    $\mathbf{z}_k \leftarrow \mathbf{z}_{k-1} - \eta \cdot \frac{\nabla_{\mathbf{z}} L_{\text{total}}}{\max(|\nabla_{\mathbf{z}} L_{\text{total}}|)}$ 
9:   for  $x_j \in \mathcal{G}$  do
10:    if  $x_h \in \text{Pa}(x_j)$  changed then
11:       $W_j \leftarrow \epsilon$ 
12:    end if
13:    if  $x_j$  changed then
14:       $W_j \leftarrow 1$ 
15:    end if
16:   end for
17:    $\mathbf{x}_k \leftarrow D(\mathbf{z}_k)$ 
18:    $\hat{y}_k \leftarrow f(\mathbf{x}_k), \hat{\mathbf{c}}_k \leftarrow f_{\text{ctx}}(\mathbf{z}_k)$ 
19:    $L_{\text{ctx}} \leftarrow \mathcal{L}_{\text{ctx}}(\hat{\mathbf{c}}_k, \mathbf{c}_{\text{cf}})$ 
20:    $L_{\text{total}} \leftarrow \mathcal{L}_{\text{total}}(\mathbf{x}_0, \mathbf{x}_k, \mathbf{z}_k, \hat{y}_k, \hat{\mathbf{c}}_k, \mathbf{W}^{\text{causal}})$ 
21:   if  $\hat{y}_k \geq \tau$  then
22:      $\mathbf{x}_{\text{cf}} \leftarrow \mathbf{x}_k$ 
23:      $\mathbf{z}_{\text{cf}} \leftarrow \mathbf{z}_k$ 
24:      $L_{\text{total}}^{\text{best}} \leftarrow L_{\text{total}}$ 
25:   end if
26:   if  $\hat{y}_k \geq \tau$  and  $L_{\text{ctx}} \leq L_{\text{ctx}}^{\text{best}}$  and  $L_{\text{total}} \leq L_{\text{total}}^{\text{best}}$  then
27:      $\mathbf{x}_{\text{cf}} \leftarrow \mathbf{x}_k$ 
28:      $\mathbf{z}_{\text{cf}} \leftarrow \mathbf{z}_k$ 
29:      $L_{\text{total}}^{\text{best}} \leftarrow L_{\text{total}}$ 
30:      $L_{\text{ctx}}^{\text{best}} \leftarrow L_{\text{ctx}}$ 
31:   end if
32: end for

```

Algorithm 3 CausalCACTUS counterfactual generation

4.2.2 Causal Feasibility Constraints

To ensure causally consistent counterfactual generation, we introduce a dynamic feasibility mechanism that constrains parent–child dependencies derived from the learned DAG. Intuitively, causally dependent nodes (i.e., child features) are dynamically unlocked whenever changes occur in their ancestor nodes (i.e., parent features). Given the learned DAG $\mathcal{G}$, we first extract the set of parent–child relationships $(x_h \rightarrow x_j)$, where the objective is to impose *causal feasibility constraints* through a causal distance loss that respects these dependencies. Algorithm 3 illustrates pseudo-codes for the counterfactual generation process.

Causal distance loss. We construct an adaptive binary causal mask $\mathbf{W}^{\text{causal}} \in \mathbb{R}^n$, initialized as $\mathbf{W}^{\text{causal}} = (1, \dots, 1)$ with length n (Algorithm 3, Line 2). The mask is updated at each iteration to constrain or encourage changes in specific feature dimensions. In other words, the mask constrains infeasible feature modifications while allowing child features to change once their parents have been modified. To achieve this, we modify the distance loss (Eq. 2) as:

$$\mathcal{L}_{\text{dist}}^{\text{causal}}(x_0, \underline{\mathbf{x}}, \mathbf{W}^{\text{causal}}) = \|\mathbf{W}^{\text{causal}} \odot (x_0 - \underline{\mathbf{x}})\|_2, \quad (7)$$

where $\odot$ denotes element-wise multiplication and $\mathbf{W}^{\text{causal}}$ is updated at each iteration. Let $\Delta x_h^{(k)} = x_h^{(k)} - x_h^{(0)}$ denote the feature change for $x_h^{(k)}$ at the k -th iteration. For each child node x_j with parent set $\text{Pa}(x_j)$, we update:

$$W_j^{\text{causal}} = \begin{cases} \epsilon, & \text{if } \exists x_h \in \text{Pa}(x_j) \text{ such that } \Delta x_h^{(k)} \neq 0, \\ 1, & \text{otherwise,} \end{cases} \quad (8)$$

where ϵ is a weight score close to 0 (instantiated as $\epsilon = 0.01$ in our experiments) that lowers the distance loss for child features once a parent has changed, hence encouraging change to happen. The causal mask update is shown in Algorithm 3, Line 9–16.

With the modified $\mathcal{L}_{\text{dist}}^{\text{causal}}$, the total loss in Eq. 4 is reformulated as below²:

$$\begin{aligned} \mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\mathbf{y}}, \hat{\mathbf{c}}, \mathbf{W}^{\text{causal}}) &= \alpha \mathcal{L}_{\text{val}}(\underline{\mathbf{y}}, y_{\text{cf}}) + (1 - \alpha) \mathcal{L}_{\text{ctx}}(\hat{\mathbf{c}}, c_{\text{cf}}) \\ &\quad + \gamma \mathcal{L}_{\text{dist}}^{\text{causal}}(x_0, \underline{\mathbf{x}}, \mathbf{W}^{\text{causal}}), \end{aligned} \quad (9)$$

where α controls the trade-off between prediction validity and contextual validity, while γ defines the strength of causal feasibility constraints. The updated loss L_{total} from Eq. 9 is used during latent optimization (illustrated in Fig. 3b), encouraging generated counterfactuals that are both context-aware and causally consistent.

In addition to causal dependencies, we explicitly incorporate feature-range clipping into feasibility constraints through iterative operations during counterfactual optimization. After each update and decoding step, the generated counterfactual $\underline{\mathbf{x}} = D(\underline{\mathbf{z}})$ is clipped using valid feature ranges:

$$\underline{\mathbf{x}} = \text{clip}(\underline{\mathbf{x}}, [x_{\min}, x_{\max}]) \quad (10)$$

where $x_{\min}$ and $x_{\max}$ denote lower and upper bounds derived from training data $\mathcal{D}_{\text{train}}$ (Line 6). Specifically, the clipping step is applied before causal mask updates, such that all counterfactual changes remain plausible and causal feasibility constraints are evaluated only on valid feature changes.

²The expression $\mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\mathbf{y}}, \hat{\mathbf{c}}, y_{\text{cf}}, c_{\text{cf}}, \alpha, \gamma, \mathbf{W}^{\text{causal}})$ is simplified as $\mathcal{L}_{\text{total}}(x_0, \underline{\mathbf{x}}, \underline{\mathbf{z}}, \underline{\mathbf{y}}, \hat{\mathbf{c}}, \mathbf{W}^{\text{causal}})$ throughout the manuscript to improve the readability of the notation.

4.2.3 Phase-Based Optimization

To improve optimization stability in CausalCACTUS, we introduce a *phase-based optimization* strategy that decomposes counterfactual generation into sequential steps in selecting the final counterfactuals. More specifically, it is divided as follows:

Phase 1: validity optimization. The objective is to obtain a valid counterfactual, i.e., a counterfactual $\underline{x}$ with predicted probability exceeding a predefined threshold τ , i.e., $f(\underline{x})_{y_{cf}} \geq \tau$ (Algorithm 3, Line 21–25). The first valid counterfactual encountered is stored as a candidate solution. During this phase, the algorithm prioritizes validity while allowing greater flexibility in contextual alignment and distance minimization.

Phase 2: context-aware and loss optimization. After a valid counterfactual is selected, optimization proceeds to the second phase, where the goal is to improve context validity and total losses while preserving predictive validity (Line 26–31). Specifically, among valid counterfactuals, we update the counterfactual candidate if either: (1) the predicted context matches the target context, or (2) the contextual loss L_{ctx} is reduced; while simultaneously requiring a decrease compared to the previous best total loss, i.e., $L_{total} \leq L_{total}^{best}$.

To balance objectives, we propose an adaptive strategy for α in Eq. 9. Before Phase 1 (i.e., validity optimization), α is set to 0.99; after Phase 1, α is progressively reduced to 0.7, increasing the relative importance of context validity and causal distance. After Phase 2, $\alpha = 0.6$, such that generated counterfactuals are more balanced to the context validity. This strategy is effective for high-dimensional latent space optimization, where early iterations might be dominated by the classifier's gradients.

5 Experimental Setups

5.1 Data Preparation

The proposed CausalCACTUS method was evaluated on five publicly available tabular datasets across different application domains, summarized in Table 1. Three of these datasets focus on credit approval prediction: Give Me Credit (*Credit*) (Hayashi, 2022), German Credit (*German*) (Hofmann, 1994), and Home Equity Line of Credit (*HELOC*) (FICO: Explainable Machine Learning, 2018). The prediction target is a binary credit risk (or loan outcome) label, and each instance corresponds to an individual described by financial and demographic attributes. Additionally, we considered two datasets from other domains: *Adult* (Becker & Kohavi, 1996) income dataset with the prediction target as income levels, and *Law* (Wightman, 1998) from education, where the outcome is bar passage for law school

Table 1 Summary of datasets used in our experiments, including sample size, feature composition (number of numerical/categorical features), context variables, and prediction target

Dataset	#Samples	#Feat (#Num/#Cat)	Context	Target
Credit	16,668	8 (8/0)	Age (>50), Dependents (≥ 1)	Delay risk
Adult	15,016	7 (4/3)	Sex, Race	Income
Law	3,672	9 (8/1)	Sex, Race	Pass bar
German	462	6 (2/4)	Sex, Age (>40)	Loan risk
HELOC	10,000	18 (18/0)	Credit history indicators (2)	Loan risk

students. These datasets range in sample size from 462 to 16,668 instances and include varying numbers of features (both categorical and numerical), which demonstrate the generalizability of our experiments.

In addition to the binary label as the prediction outcome, we derived context variables from each dataset as follows. For the Credit dataset, we defined two binary context variables c_i from demographic features: Age ($c_1 = 0: \leq 50, c_1 = 1: >50$) and Number of Dependents ($c_2 = 0: \text{none}, c_1 = 1: \geq 1$ dependents). Similarly, we chose socio-demographic features (i.e., Sex, Race, and Age) to define binary indicators as context variables for Adult, Law, and German datasets. For the HELOC dataset, we used two features indicating credit history length as context variables to represent user-specific financial history. All datasets were scaled and standardized before model training and counterfactual generation; categorical features were encoded using one-hot representations. Instances with missing values were discarded during data preprocessing. All experiments are conducted using five random repeated tests with an 80/20% stratified train-test split across all datasets. For classifier and AE model training, 15% of the training data is split as a holdout validation set for early stopping. For counterfactual evaluation, 50 randomly selected test instances are evaluated in each run.

5.2 Baseline Models

CausalCACTUS was compared against several counterfactual generation methods using latent space representations, including LatentCF++ (Wang et al., 2024, 2021) and PrototypeCF (Van Looveren & Klaise, 2021). For PrototypeCF, we compared two variants: Prototype-C, which incorporated the AE model's reconstruction error, and Prototype-D, which did not. The original CACTUS method (Pérez et al., 2025) was also included in the baseline models.

5.3 Modeling Setups

The classifier and β -VAE models in the proposed framework were trained prior to counterfactual generation. For all tabular datasets in the experiments, we considered a standard DNN classifier composed of four fully connected layers and an output layer with softmax activation that outputs the positive class probability. For the β -VAE used for counterfactual generation, we trained an encoder consisting of two fully connected layers that map input instances into a 4-dimensional latent space. The decoder reconstructs the input from the latent representation using two fully connected layers. Finally, the context output from the VAE model was estimated from the latent representation using a fully connected layer with sigmoid activation to estimate the context vector. The AE models for baseline comparisons were trained using three fully connected layers with ReLU activation functions. Early stopping was applied with a patience of 20.

The DAG for each dataset was learned using the ReX method (Renero et al., 2026); hyperparameters were set to their default values using the implementation from the supporting website³. Additionally, we evaluated two levels of structural prior knowledge for each dataset. For example, the following priors for the Credit dataset were incorporated into the CausalCACTUS evaluation:

³<https://github.com/renero/causalexplain/>.

- *Minimal priors*: context variables were constrained to be root causes, while the other variables formed the second-tier hierarchy (i.e., two tiers of priors).
- *Moderate priors*: variables were separated into context, financial capacity, and credit behavior tiers (i.e., three tiers of priors).

Similar two-level prior setups were used when learning DAGs for the other four datasets. For each dataset, two DAGs (one per prior level) were learned; and all learned DAGs are available as supplementary materials in Appendix A.

For the first experiment, we evaluated the proposed CausalCACTUS against baseline models for the context-preserving scenario. The proposed CausalCACTUS was run with a dynamic α with the phase-based optimization mechanism (gradually decreasing from 0.99 to 0.6), along with $\gamma = 0.1$. The original CACTUS was evaluated with $\alpha = 0.7$, $\gamma = 0.1$, and maximum epochs $T = 300$. The PrototypeCF and LatentCF++ models were evaluated using the default settings specified in their respective papers (Van Looveren & Klaise, 2021; Wang et al., 2021). For the ablation study, we first investigated the effect of α by varying it from 0.01 to 0.99 in approximate steps of 0.25, while fixing all other hyperparameters in both CACTUS and CausalCACTUS. Additionally, we compared the original CACTUS method with variants of combining two proposed augmented components in CausalCACTUS: (1) causal feasibility constraints, and (2) phase-based optimization. All experiments are evaluated on a Tesla V100 DGXS 32GB GPU and an Intel(R) Xeon(R) CPU E5-2698 40-Core Processor.

5.4 Evaluation Metrics

The counterfactual examples generated by CausalCACTUS and the baseline models are quantitatively evaluated using the following standard metrics from the literature:

- *Validity* (Mahajan et al., 2019; Wang & Bilgic, 2023), defined as the accuracy score between the predicted label $\hat{y}_{\text{cf,label}}^{(i)}$ and the desired target label $y_{\text{cf}}^{(i)}$:

$$\text{Validity} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{1} \left(\hat{y}_{\text{cf,label}}^{(i)} = y_{\text{cf}}^{(i)} \right),$$

where N_{test} is the number of test instances. Higher validity is better.

- *Compactness* (Karlsson et al., 2018; Wang et al., 2024), similar to sparsity, measured as the proportion of features that differ between the counterfactual x_{cf} and the original instance x_0 , relative to the total number of input features n :

$$\text{Compactness} = \frac{1}{N_{\text{test}} \cdot n} \sum_{i=1}^{N_{\text{test}}} \sum_{j=1}^n \mathbb{1} \left(|x_j^{(i)} - x_{\text{cf},j}^{(i)}| \leq \text{atol} \right),$$

where atol is a small tolerance. A higher compactness score is desired.

- *Proximity* (Laugel et al., 2019; Mahajan et al., 2019; Pawelczyk et al., 2020), computed as the normalized Euclidean distance between the counterfactual x_{cf} and the original instance x_0 , normalized by the square root of the number of features n , i.e.,

$$Proximity = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} \frac{\|x^{(i)} - x_{cf}^{(i)}\|_2}{\sqrt{n}}.$$

Lower proximity indicates fewer modifications from the original samples and is better.

- *Context-LOF_k*, similar to the LOF metric (Kanamori et al., 2020), measures whether a counterfactual lies within the data distribution of its target context. The neighborhood size k is set to the square root of n , with a minimum of 20 and a maximum of 100. For each target context c_{cf} , we train a LOF model on $\mathcal{D}_{train}^{(c)} = \{x^{(i)} \in \mathcal{D}_{train} : c^{(i)} = c_{cf}\}$. Then, Context-LOF is computed as the average of negative LOF scores for x_{cf} in the target context c_{cf} :

$$Context-LOF = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} -LOF(x_{cf}^{(i)}).$$

Lower values (i.e., LOF scores close to 1) indicate better in-distribution alignment with the target context.

- *Causal-compact Consistency*: we extend causal-constraint validity (CCV) (Duong et al., 2024) by combining it with compactness. Given the DAG $\mathcal{G} = (V, \mathcal{E})$ and parent sets $Pa(j)$ for each feature x_j , we calculate the feature change as $\Delta x_j^{(i)} = |x_{cf,j}^{(i)} - x_j^{(i)}|$. CCV is then computed as the average fraction of satisfied parent–child constraints:

$$CCV = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} \frac{1}{|\mathcal{E}|} \sum_{j=1}^{|\mathcal{E}|} 1 \left(\Delta x_j^{(i)} = 0 \text{ or } \exists h \in Pa(j) \text{ s.t. } \Delta x_h^{(i)} \neq 0 \right),$$

where $\mathcal{E}$ denotes the set of directed edges in $\mathcal{G}$ and x_h represents the parent node of x_j . The final metric is calculated as:

$$Causal-compact Consistency = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} CCV^{(i)} \times Compactness^{(i)}.$$

Higher values indicate sparser counterfactual changes that are also causally consistent.

6 Empirical Results

6.1 Quantitative Results

Table 2 reports quantitative results comparing CausalCACTUS with baseline models in counterfactual generation, based on five runs of 50 test instances each. Results are summarized as the mean and standard deviation for each evaluation metric. All counterfactuals are generated under the context-preserving setting (i.e., $c_{cf} = c_0$). Across datasets and prior settings, CausalCACTUS ranked among the top two validity scores in 8 out of 10 comparisons, closely following the LatentCF++ and PrototypeCF-based models. Specifically, CausalCACTUS substantially improved validity compared to the original CACTUS method, increasing from approximately 0.7 to around 0.9 across comparisons. This result suggests that the augmented components in CausalCACTUS enhance its effectiveness and convergence stability in generating valid counterfactuals.

In terms of context-LOF, CausalCACTUS outperformed all baseline models in 9 out of 10 comparison scenarios, with the original CACTUS mostly ranking second. This highlights the ability of CausalCACTUS to better preserve the original context using causal graphs as constraints. For both compactness and proximity, CausalCACTUS consistently achieved the best performance across most datasets. The exception was proximity on the German dataset, where it ranked second; while CACTUS generally obtained second-best scores across these two metrics. The augmented CausalCACTUS improved these two metrics of the original version by about 0.1–0.2, indicating that the augmentation effectively encouraged sparser changes that follow causal feasibility constraints. This comparison highlights a trade-off between prediction validity and generating compact, proximate counterfactuals, indicating that CausalCACTUS achieved a favorable balance. The trade-off between prediction and context validity was further investigated in the ablation study.

Regarding causal-compact consistency, CausalCACTUS mostly achieved the highest score in 9 out of 10 experimental configurations, outperforming all other baselines. In comparison, CACTUS ranked second in many comparisons. This metric jointly captures the proportions of counterfactual changes that are both causally coherent and minimal, where higher scores from CausalCACTUS indicate stronger causal consistency with minimal perturbations. The low variance across experimental runs further suggests these improvements are stable and can generalize to multiple dataset comparisons. Furthermore, the effects of the augmented causal feasibility and phase-based components in CausalCACTUS were analyzed in the ablation study.

6.2 Ablation Studies

First, we performed an ablation study to examine the effects of the hyperparameter α (defined in Eq. 9) in the counterfactual search of CausalCACTUS, compared to the original CACTUS, to explore the interactions between context and validity loss. Figure 4 reports the mean and standard deviation of the validity, context-LOF, and compactness metrics across five datasets over five repeated experimental runs for CausalCACTUS (under both the moderate and minimum priors) and CACTUS. From the figure, we observed a consistent upward trend in both validity and context-LOF as α increases, across all datasets and both prior settings. Moreover, the CausalCACTUS model exhibited a lower standard deviation

Table 2 Quantitative comparison of CausalCACTUS and baseline methods across datasets under minimal (Min.) and moderate (Mod.) prior settings, evaluated using validity, context-LOF, compactness, proximity, and causal-compact consistency

Prior	Dataset	Model	Valid.	Con-LOF	Compa.	Proxi.	Cau-com.
Min.	Credit	CACTUS	0.72±0.03	<u>1.27±0.02</u>	<u>0.10±0.01</u>	<u>0.55±0.05</u>	0.38±0.03
		CausalCAC	<u>0.99±0.01</u>	1.21±0.04	0.18±0.02	0.53±0.04	0.38±0.03
		LatentCF++	1.00±0.00	1.50±0.10	0.04±0.01	0.66±0.06	<u>0.34±0.02</u>
		Prototype C	1.00±0.00	3.72±0.21	0.01±0.00	1.83±0.05	0.21±0.02
		Prototype D	1.00±0.00	2.19±0.08	0.01±0.01	1.29±0.06	0.26±0.02
	Adult	CACTUS	0.55±0.06	<u>1.11±0.02</u>	<u>0.13±0.03</u>	0.63±0.02	<u>0.38±0.02</u>
		CausalCAC	<u>0.81±0.02</u>	1.10±0.03	0.35±0.02	0.55±0.03	0.51±0.01
		LatentCF++	1.00±0.00	1.17±0.04	0.11±0.01	<u>0.61±0.11</u>	0.33±0.05
		Prototype C	1.00±0.00	2.13±0.06	0.03±0.01	2.71±0.10	0.07±0.01
		Prototype D	1.00±0.00	1.81±0.08	0.04±0.01	2.46±0.08	0.08±0.01
	Law	CACTUS	0.61±0.05	1.05±0.01	<u>0.06±0.02</u>	<u>0.49±0.03</u>	<u>0.18±0.02</u>
		CausalCAC	<u>0.81±0.04</u>	1.05±0.01	0.27±0.04	0.35±0.04	0.33±0.04
		LatentCF++	1.00±0.01	<u>1.11±0.02</u>	0.03±0.01	0.51±0.07	0.17±0.00
		Prototype C	0.87±0.06	2.90±0.13	0.01±0.01	2.47±0.17	0.06±0.01
		Prototype D	0.97±0.01	2.09±0.09	0.02±0.01	1.86±0.07	0.09±0.01
	German	CACTUS	<u>0.86±0.04</u>	1.03±0.01	0.02±0.01	0.61±0.01	0.26±0.01
		CausalCAC	0.85±0.04	<u>1.04±0.01</u>	0.17±0.03	<u>0.48±0.02</u>	0.40±0.02
		LatentCF++	1.00±0.00	1.19±0.02	<u>0.03±0.01</u>	0.38±0.05	<u>0.38±0.04</u>
		Prototype C	1.00±0.00	3.00±0.10	0.01±0.01	2.22±0.07	0.09±0.01
		Prototype D	1.00±0.00	1.99±0.08	0.02±0.00	1.53±0.04	0.13±0.02
	HELOC	CACTUS	0.52±0.06	1.21±0.08	<u>0.06±0.03</u>	0.43±0.03	<u>0.22±0.03</u>
		CausalCAC	<u>0.81±0.03</u>	1.21±0.08	0.22±0.03	0.38±0.03	0.32±0.03
		LatentCF++	1.00±0.00	<u>1.33±0.08</u>	0.03±0.01	<u>0.39±0.06</u>	0.20±0.02
		Prototype C	1.00±0.00	2.94±0.13	0.03±0.00	1.72±0.05	0.06±0.01
		Prototype D	1.00±0.00	2.00±0.12	0.02±0.00	1.26±0.07	0.05±0.01
Mod.	Credit	CACTUS	0.73±0.06	<u>1.28±0.06</u>	<u>0.11±0.04</u>	<u>0.57±0.04</u>	0.48±0.04
		CausalCAC	<u>0.98±0.02</u>	1.24±0.07	0.15±0.03	0.56±0.04	<u>0.46±0.02</u>
		LatentCF++	1.00±0.00	1.55±0.10	0.04±0.01	0.72±0.08	0.40±0.01
		Prototype C	1.00±0.00	3.79±0.18	0.01±0.00	1.89±0.05	0.22±0.02
		Prototype D	1.00±0.00	2.31±0.13	0.02±0.01	1.29±0.06	0.31±0.02
	Adult	CACTUS	0.63±0.10	<u>1.13±0.02</u>	0.10±0.05	<u>0.62±0.11</u>	<u>0.36±0.04</u>
		CausalCAC	<u>0.72±0.09</u>	1.09±0.02	0.42±0.08	0.41±0.08	0.56±0.07
		LatentCF++	1.00±0.00	1.16±0.02	<u>0.11±0.01</u>	0.63±0.09	0.31±0.02
		Prototype C	1.00±0.00	2.12±0.11	0.03±0.01	2.74±0.12	0.07±0.01
		Prototype D	1.00±0.00	1.80±0.03	0.03±0.01	2.52±0.04	0.07±0.01
	Law	CACTUS	0.63±0.04	<u>1.05±0.01</u>	<u>0.05±0.02</u>	0.51±0.01	0.14±0.02
		CausalCAC	0.85±0.04	1.04±0.01	0.23±0.04	0.38±0.04	0.27±0.04
		LatentCF++	0.99±0.01	1.14±0.03	0.03±0.01	<u>0.50±0.08</u>	<u>0.15±0.01</u>
		Prototype C	0.88±0.02	2.88±0.09	0.01±0.01	2.43±0.09	0.05±0.01
		Prototype D	<u>0.96±0.02</u>	2.03±0.05	0.02±0.00	1.81±0.10	0.06±0.01
	German	CACTUS	0.80±0.02	1.04±0.01	0.02±0.01	0.55±0.02	<u>0.38±0.03</u>
		CausalCAC	<u>0.83±0.01</u>	1.04±0.01	0.19±0.01	<u>0.41±0.02</u>	0.52±0.02
		LatentCF++	1.00±0.00	<u>1.18±0.02</u>	<u>0.03±0.01</u>	0.36±0.02	0.52±0.02
		Prototype C	1.00±0.00	3.09±0.16	0.01±0.01	2.18±0.07	0.10±0.01
		Prototype D	1.00±0.00	2.13±0.12	0.02±0.01	1.54±0.05	0.12±0.01
	HELOC	CACTUS	0.54±0.12	1.17±0.07	<u>0.06±0.02</u>	0.45±0.03	<u>0.22±0.02</u>

Table 2 (continued)

Prior	Dataset	Model	Valid.	Con-LOF	Compa.	Proxi.	Cau-com.
		CausalCAC	<u>0.80 ± 0.03</u>	1.17 ± 0.07	0.22 ± 0.03	<u>0.37 ± 0.02</u>	0.34 ± 0.02
		LatentCF++	1.00 ± 0.00	1.31 ± 0.08	0.03 ± 0.00	0.40 ± 0.03	<u>0.22 ± 0.01</u>
		Prototype C	1.00 ± 0.00	2.89 ± 0.12	0.03 ± 0.00	1.72 ± 0.03	0.06 ± 0.01
		Prototype D	1.00 ± 0.00	2.02 ± 0.05	0.02 ± 0.01	1.26 ± 0.05	0.05 ± 0.01

Reported scores represent the mean $\pm$ standard deviation over five experimental runs. For each dataset and metric, the best and second-best results are highlighted in bold and underlined, respectively

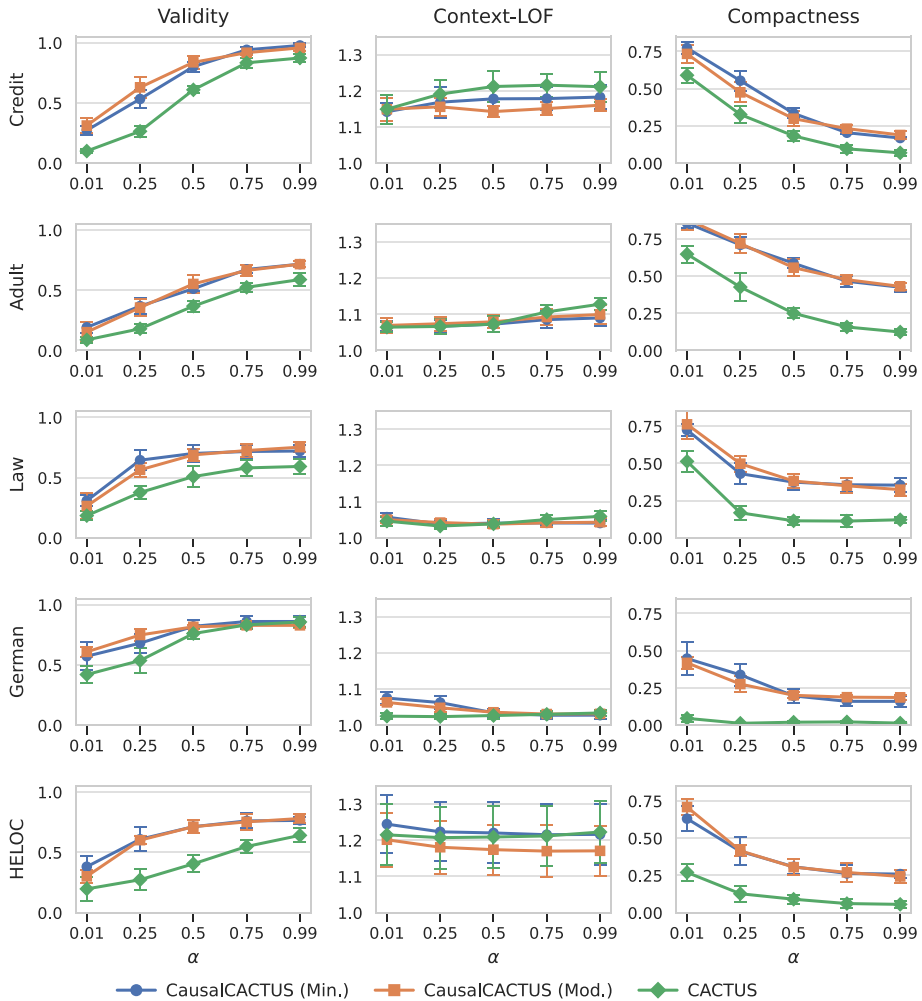


Fig. 4 Effect of the hyperparameter α illustrating the trade-off between validity, context-LOF, and compactness for the original CACTUS model and CausalCACTUS under minimal (Min.) and moderate (Mod.) prior settings across five datasets. Each row corresponds to a dataset, and each column reports a metric as α increases gradually from 0.01 to 0.99

tion, indicating higher stability for generating desired counterfactuals. A larger value of α also leads to lower compactness, indicating a larger proportion of feature changes. Overall, increasing α introduces a trade-off between validity and context-LOF in both CACTUS and CausalCACTUS models, with $\alpha \approx 0.7$ representing a balanced choice that achieves strong performance in both evaluation metrics. Additionally, a dynamic setup of α using the phase-based optimization was investigated in the following ablation study.

In addition, we evaluated the individual and combined effects of two augmented components from CausalCACTUS under minimal and moderate prior settings across five datasets, shown in Table 3. In the comparison, we observed that incorporating both the causal feasibility constraints (CC) and the phase-based optimization (PO) achieved the best performance in terms of proximity and causal-compact consistency in 9 out of 10 settings. Furthermore, the PO component substantially improved validity and compactness under both prior settings compared to the Base model in most comparisons, while augmenting the CC component primarily increased context-LOF and causal-compact consistency. In sum, CausalCACTUS combining both components consistently achieved the balanced overall performance, showing a favorable improvement across all five metrics under both priors.

6.3 Critical Difference Diagrams

To further assess the statistical differences of evaluation metrics, we performed a Friedman test followed by the Wilcoxon-Holm post-hoc analysis (Ismail Fawaz et al., 2019) across the ten benchmark tasks (i.e., five datasets under two prior settings). Figure 5 illustrates the critical difference diagrams showing average ranks for each counterfactual method, where lower average ranks indicate better overall performance.

For the main comparison, CausalCACTUS consistently achieved the best average rank across proximity, context-LOF, compactness, and causal-compact consistency, with average ranks ranging from 1.0 to 1.4 (see Fig. 5c, e, g, i). In terms of validity, CausalCACTUS outperformed the original CACTUS, demonstrating that the proposed augmentation with causal dependencies improves counterfactual validity. These diagrams further demonstrate that CausalCACTUS substantially outperformed other baselines in context- and causality-related metrics, while achieving a favorable trade-off between validity and other metrics compared to CACTUS. Moreover, for the ablation study (Fig. 5b, d, f, h, j), we observed that the full CausalCACTUS model (Base+CC+PO) consistently achieved the best average ranks (between 1.2 and 1.9) across all evaluation metrics. These results suggest that incorporating both causal feasibility constraints (Base+CC) and phase-based optimization (Base+PO) components in CausalCACTUS provides the most optimal performance enhancement, while both components improve the lowest-ranked CACTUS (Base) model.

6.4 Qualitative Analysis

In this section, we present two qualitative analyses to illustrate (1) the structure of counterfactual changes using two grouped feature-difference visualizations and (2) representative counterfactual examples to examine CausalCACTUS in the context-preserving and causal-constraint setting. To avoid redundancy, we discuss the Credit dataset as a representative example, while analogous qualitative analyses for the remaining datasets are provided in Appendix B.

Table 3 Ablation study of CausalCACTUS components under minimal (Min.) and moderate (Mod.) prior settings across five datasets

Prior	Dataset	Variants	Valid.	Con-LOF	Compa.	Proxi.	Cau-Com.
Min.	Credit	Base	0.72±0.03	1.27±0.02	<u>0.10±0.01</u>	0.55±0.05	<u>0.38±0.03</u>
		Base+CC	0.71±0.03	1.24±0.02	0.18±0.03	<u>0.54±0.05</u>	0.39±0.02
		Base+PO	<u>0.98±0.01</u>	<u>1.23±0.03</u>	0.08±0.02	0.53±0.05	<u>0.38±0.03</u>
		Base+CC+PO	0.99±0.01	1.21±0.04	0.18±0.02	0.53±0.04	<u>0.38±0.03</u>
	Adult	Base	0.55±0.06	<u>1.11±0.02</u>	0.13±0.03	<u>0.63±0.02</u>	0.38±0.02
		Base+CC	0.56±0.07	<u>1.11±0.01</u>	0.28±0.04	0.63±0.03	0.47±0.03
		Base+PO	<u>0.80±0.01</u>	1.10±0.02	0.22±0.02	0.55±0.03	0.44±0.02
		Base+CC+PO	0.81±0.02	1.10±0.03	0.35±0.02	0.55±0.03	0.51±0.01
	Law	Base	0.61±0.05	1.05±0.01	0.06±0.02	0.49±0.03	0.18±0.02
		Base+CC	0.60±0.06	1.05±0.01	0.13±0.02	0.49±0.03	0.19±0.02
		Base+PO	0.82±0.05	1.05±0.01	<u>0.21±0.05</u>	<u>0.36±0.04</u>	<u>0.32±0.05</u>
		Base+CC+PO	<u>0.81±0.04</u>	1.05±0.01	0.27±0.04	0.35±0.04	0.33±0.04
	German	Base	0.86±0.04	1.03±0.01	0.02±0.01	<u>0.61±0.01</u>	0.26±0.01
		Base+CC	0.84±0.02	1.03±0.01	0.03±0.01	<u>0.61±0.01</u>	<u>0.27±0.02</u>
		Base+PO	<u>0.85±0.04</u>	<u>1.04±0.01</u>	<u>0.16±0.04</u>	0.48±0.02	0.40±0.02
		Base+CC+PO	<u>0.85±0.04</u>	<u>1.04±0.01</u>	0.17±0.03	0.48±0.02	0.40±0.02
	HELOC	Base	<u>0.52±0.06</u>	1.21±0.08	0.06±0.03	<u>0.43±0.03</u>	0.22±0.03
		Base+CC	<u>0.52±0.06</u>	1.21±0.08	0.11±0.04	<u>0.43±0.04</u>	<u>0.23±0.03</u>
		Base+PO	0.81±0.03	<u>1.22±0.08</u>	<u>0.21±0.03</u>	0.38±0.03	0.32±0.04
		Base+CC+PO	0.81±0.03	1.21±0.08	0.22±0.03	0.38±0.03	0.32±0.03
Mod.	Credit	Base	0.73±0.06	1.28±0.06	0.11±0.04	0.57±0.04	0.48±0.04
		Base+CC	0.72±0.06	1.26±0.04	0.20±0.03	0.55±0.03	0.48±0.03
		Base+PO	0.98±0.03	<u>1.25±0.07</u>	0.07±0.03	<u>0.56±0.04</u>	<u>0.46±0.02</u>
		Base+CC+PO	0.98±0.02	1.24±0.07	<u>0.15±0.03</u>	<u>0.56±0.04</u>	<u>0.46±0.02</u>
	Adult	Base	0.63±0.10	1.13±0.02	0.10±0.05	0.62±0.11	0.36±0.04
		Base+CC	0.63±0.09	1.12±0.02	0.26±0.05	0.62±0.11	0.42±0.04
		Base+PO	0.73±0.09	<u>1.10±0.03</u>	<u>0.29±0.09</u>	<u>0.42±0.08</u>	<u>0.52±0.07</u>
		Base+CC+PO	<u>0.72±0.09</u>	1.09±0.02	0.42±0.08	0.41±0.08	0.56±0.07
	Law	Base	0.63±0.04	<u>1.05±0.01</u>	0.05±0.02	<u>0.51±0.01</u>	0.14±0.02
		Base+CC	0.63±0.04	<u>1.05±0.01</u>	0.12±0.03	<u>0.51±0.01</u>	<u>0.15±0.02</u>
		Base+PO	<u>0.84±0.04</u>	<u>1.05±0.01</u>	<u>0.19±0.05</u>	0.38±0.04	0.27±0.04
		Base+CC+PO	0.85±0.04	1.04±0.01	0.23±0.04	0.38±0.04	0.27±0.04
	German	Base	0.80±0.02	<u>1.04±0.01</u>	0.02±0.01	0.55±0.02	0.38±0.03
		Base+CC	0.79±0.01	1.03±0.01	<u>0.03±0.01</u>	0.55±0.02	<u>0.39±0.03</u>
		Base+PO	0.83±0.01	<u>1.04±0.01</u>	0.19±0.01	<u>0.42±0.02</u>	0.52±0.03
		Base+CC+PO	0.83±0.01	<u>1.04±0.01</u>	0.19±0.01	0.41±0.02	0.52±0.02
	HELOC	Base	0.54±0.12	<u>1.17±0.07</u>	0.06±0.02	0.45±0.03	0.22±0.02
		Base+CC	0.54±0.12	1.16±0.07	0.09±0.03	0.45±0.04	0.22±0.03
		Base+PO	0.81±0.03	<u>1.17±0.06</u>	<u>0.21±0.03</u>	<u>0.38±0.03</u>	<u>0.33±0.02</u>
		Base+CC+PO	<u>0.80±0.03</u>	<u>1.17±0.07</u>	0.22±0.03	0.37±0.02	0.34±0.02

Variants denote phase-based optimization (PO) and causal feasibility constraints (CC) to the original CACTUS (Base); CausalCACTUS is shown as Base+PO+CC. Results are reported as mean ± standard deviation over five repeated tests across five datasets. The best and second-best results are highlighted in bold and underlined, respectively

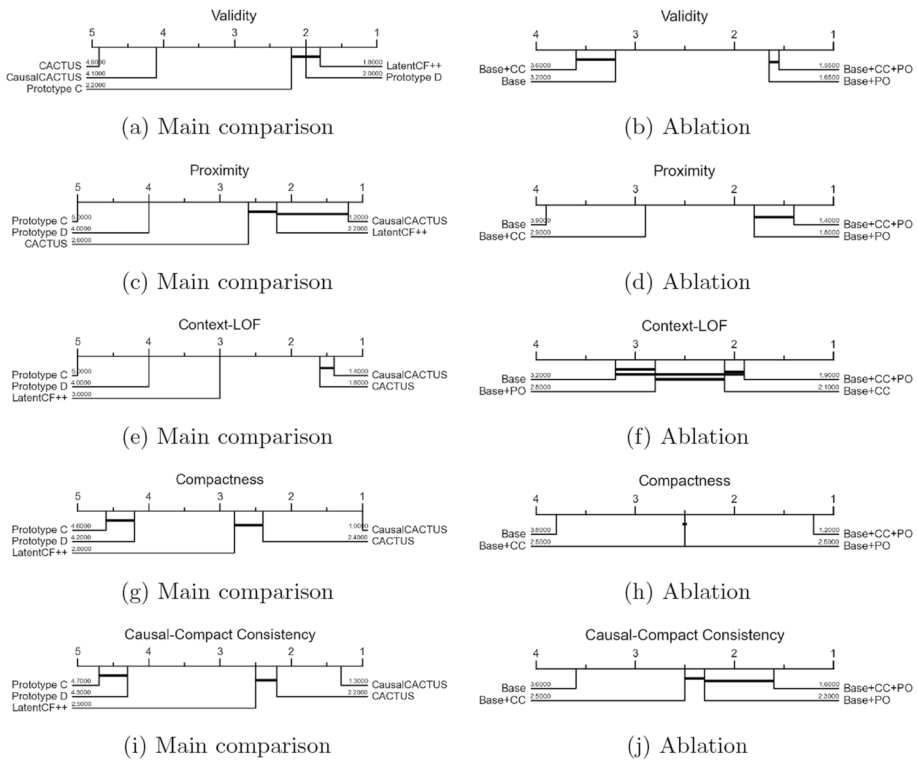


Fig. 5 Critical difference diagrams using the Wilcoxon-Holm post-hoc analysis across five datasets under two prior settings. **a, c, e, g, and i** compare CausalCACTUS with other baseline counterfactual methods, while **b, d, f, h, and j** show the ablation study comparing different CausalCACTUS variants. From top to bottom, the rows correspond to validity, proximity, context-LOF, compactness, and causal-compact consistency, respectively

Figure 6 visualizes the normalized mean absolute feature changes Δx using a heatmap, aggregated over 50 test instances and reported for each model. This figure captures the average magnitude of each individual feature perturbation. We observed that the CausalCACTUS model consistently produced smaller changes than CACTUS and LatentCF++, indicating improved proximity; while Prototype-C and Prototype-D showed more widespread modifications, indicating that their generated counterfactuals were less proximate. Overall, this analysis supports the hypothesis that CausalCACTUS generates counterfactuals with higher compactness and remains closer to the original instances.

In addition, Fig. 7a presents pairwise Spearman correlations $|\rho_s(\Delta x_i, \Delta x_j)|$ between feature changes across all test instances, where i and j index denotes different input features. We observed that Prototype-C and Prototype-D models exhibited generally weak correlations for the black-outlined cells, suggesting largely independent feature changes. In contrast, the CausalCACTUS model showed strong correlations concentrated on a small number of feature pairs, closely matching the provided causal structure (Fig. 7b). Although LatentCF++ also shows strong correlations, these are less consistently aligned with underlying causal dependencies (i.e., not overlapping with black-outlined cells). In sum, these

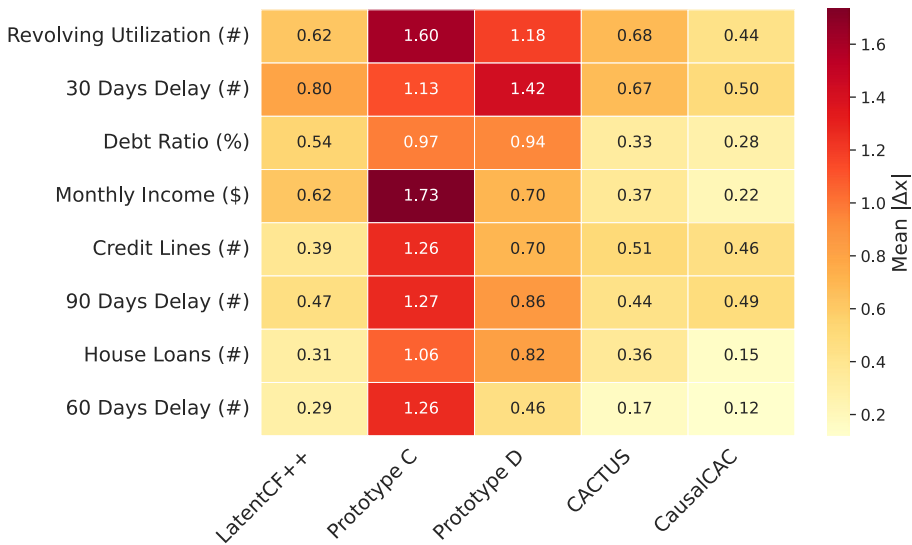
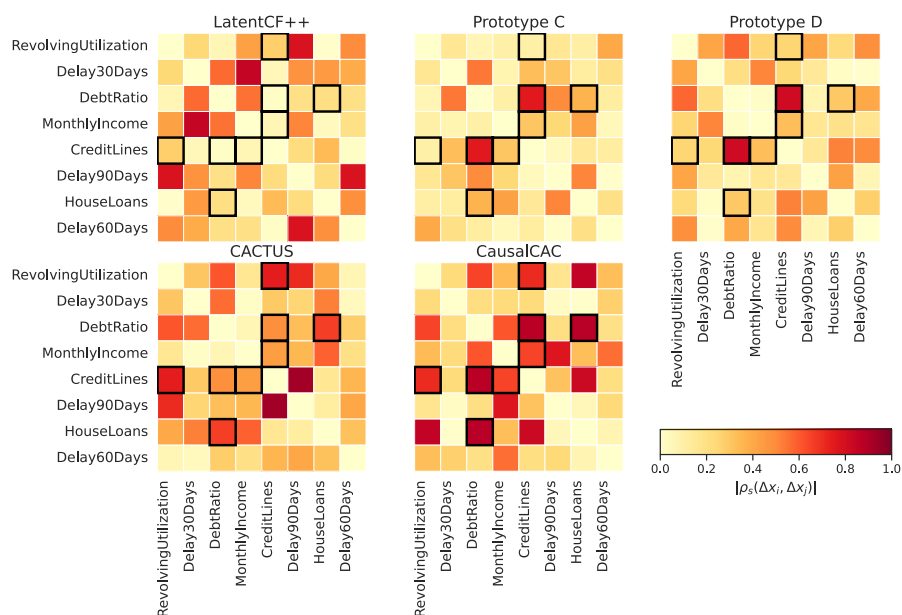


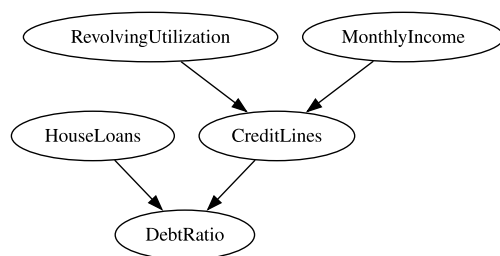
Fig. 6 Normalized mean absolute feature changes across CausalCACTUS and baseline models on the Credit dataset. Darker colors indicate larger average perturbations, highlighting differences in compactness and proximity across methods

results indicate that CausalCACTUS generates counterfactual changes that are more causally consistent with causal dependencies as constraints.

Figure 8 presents a concrete counterfactual example for a loan applicant over 50 years old (i.e., the specified context) who is initially classified as high risk of default. The figure compares the counterfactuals generated by CausalCACTUS and baseline methods under the context-preserving setting. CausalCACTUS produced more actionable and realistic counterfactuals than the PrototypeCF-based approaches. In particular, Prototype-C suggests a decrease in monthly income of 15,409 USD, which is implausible in real-world scenarios. On the other hand, Prototype-D, LatentCF++, and CACTUS proposed income increases exceeding 5,000 USD, which might be unrealistic for an applicant older than 50, given typical income conditions at this life stage. In contrast, CausalCACTUS proposed a moderate adjustment of 1,605 USD, which would be more plausible and consistent with observed socioeconomic patterns of older individuals. Besides, CausalCACTUS required fewer overall feature modifications in this example, whereas PrototypeCF-based models and LatentCF++ introduced a larger number of simultaneous changes, reducing the actionability. According to the dependencies in the learned causal graph (see Fig. 7b), the following relationship should be respected: $\text{RevolvingUtilization} \rightarrow \text{CreditLines} \rightarrow \text{DebtRatio}$. However, these causal relationships were violated by LatentCF++ and PrototypeCF-based models; but they were followed by counterfactuals generated using CausalCACTUS. Overall, this example illustrates how CausalCACTUS generates counterfactual explanations that are not only valid but also context-aware, causally consistent, and practically actionable in real-world scenarios.



(a) Pairwise Spearman correlations of normalized feature changes across methods on the Credit dataset. Black-outlined cells indicate parent-child pairs from the causal graph, where stronger dependencies are expected.



(b) Credit dataset causal graph used to define parent-child feature pairs for comparison.

Fig. 7 Correlation structure of feature-change magnitudes (a) alongside the corresponding causal graph (b) for the Credit dataset

7 Discussion

Our proposed CausalCACTUS jointly addresses counterfactual validity, proximity, context preservation, and causal consistency in a unified optimization framework. As shown in Table 2, combining phase-based optimization, causal feasibility constraints, and context-aware latent optimization provides a favorable balance among these objectives, outperforming all baselines (Pérez et al., 2025; Van Looveren & Klaise, 2021; Wang et al., 2021). The ablation study in Table 3 and Fig. 4 further indicates that the phase-based optimization primarily improves counterfactual validity by guiding the search toward the target class, while the causal feasibility constraints enhance causal consistency and context preserva-

Revolving Utilization (#)	1	-1	-1	-1	-1	-1
30 Days Delay (#)	0	+0	-1	+0	+0	+0
Debt Ratio (%)	0	+0	+1	+0	+0	+0
Monthly Income (\$)	0.36	+0.14	-0.33	-0.14	-0.10	-0.06
Credit Lines (#)	1.00	+0.00	+0.00	-1.00	+0.00	+0.00
90 Days Delay (#)	8,300	+8,376	-15,409	+5,739	+12,033	+1,605
House Loans (#)	13	+1	-4	-2	-6	-6
60 Days Delay (#)	1	+0	+0	-1	+1	+0
x_0 (original) Δ LatentCF++ Δ Prototype C Δ Prototype D Δ CACTUS Δ CausalCAC						

Fig. 8 Example counterfactual explanations from the Credit dataset. The individual is over 50 years old (context) and is initially classified as high risk of default. The first column reports the original feature values x_0 , while subsequent columns show the feature changes Δx proposed by each model under the context-preserving setting

tion, together with choosing the optimal hyperparameter α to reach a balanced performance. Together, these augmented components in CausalCACTUS improve the performance compared to the original CACTUS (Pérez et al., 2025), particularly in context-LOF and causal-compact consistency.

The qualitative analyses further support these quantitative findings. Across all five datasets, the feature-change heatmaps (Figs. 6 and 14) consistently show that CausalCACTUS modifies fewer features in smaller scales, consistent with its competitive proximity and compactness scores. Similarly, Figs. 7a and 15, 16, 17, 18 reveal stronger correlation of feature changes among causally related feature pairs, suggesting that the generated counterfactuals better reflect the underlying causal structure. Moreover, representative examples spanning credit, income, and education domains (Figs. 8 and 19, 20, 21, 22) illustrate targeted and causally consistent counterfactual changes, providing evidence that the proposed framework is applicable across different tabular classification settings. Nevertheless, further evaluation is required before extending these conclusions to other data modalities such as images and time series.

Despite these promising results, CausalCACTUS has two important limitations. First, the generated counterfactuals depend on both the estimated causal graph. Since the proposed framework assumes only a learned DAG rather than a fully specified structural causal model, CausalCACTUS relies on the DAG learned by the causal discovery ReX method (Renero et al., 2026). For example, missing edges in the estimated graph may therefore introduce inaccurate or incomplete causal dependencies into the counterfactual search. This limitation could be mitigated when reliable domain knowledge is available; expert-defined causal graphs can replace or refine automatically learned graphs to improve causal fidelity. Second, the generative model constitutes an inherent bottleneck of the proposed framework. CausalCACTUS performs the counterfactual search in a β -VAE latent space, under the assumption of improved latent representation disentanglement of the input compared to standard AE models. However, reconstruction errors or misalignment between the latent representation and the underlying causal structure may lead to information loss or unin-

tended feature changes, thereby reducing counterfactual fidelity and actionability (Wang et al., 2024).

These limitations motivate several directions for future work. Rather than relying solely on a β -VAE, future work could investigate generative backbones that provide improved reconstruction and disentanglement capabilities, while preserving the overall CausalCACTUS workflow. Promising directions include diffusion models, normalizing flows, and causally informed VAE variants that explicitly align latent representations with causal mechanisms (Bond-Taylor et al., 2022; Shen et al., 2022). In addition, jointly learning latent representations and causal structures, or integrating uncertainty estimates for the learned causal graphs, could further improve the robustness and reliability of generated counterfactual explanations (Yang et al., 2021).

8 Conclusions

We proposed CausalCACTUS, a causally consistent and context-aware framework for counterfactual explanations that incorporates user-defined contextual alignment and causal feature dependencies via a latent-space counterfactual search. By extending the original CACTUS and integrating data-driven causal dependencies learned via the ReX causal discovery method, CausalCACTUS incorporates causal feasibility constraints and a phase-based optimization strategy to guide counterfactual search, improving the convergence stability and validity of generated counterfactuals. We evaluated the framework on five publicly available tabular datasets across credit risk assessment, income prediction, and educational domains under a context-preserving scenario setting. Quantitative results showed that CausalCACTUS consistently improved the Context-LOF and causal-compact consistency metrics over the previous CACTUS baseline, while remaining competitive with all baseline models across validity, proximity, and compactness. Ablation studies highlighted a trade-off between validity and Context-LOF, demonstrating that our hyperparameter setup achieved a balanced compromise between the two objectives. Additionally, ablation analyses showed that the combination of causal feasibility constraints and phase-based optimization in CausalCACTUS consistently led to enhanced performance across metrics and causal prior settings. Qualitative analyses further validated that CausalCACTUS generated sparse counterfactual changes that were more causally consistent, generating valid counterfactuals that are practically actionable and aligned with user-defined contextual preferences. Future work could include extending CausalCACTUS to more complex decision-making settings, such as multivariate time-series classification and sequential prediction tasks. Another promising direction is the incorporation of domain-informed constraints and novel generative models to further enhance causal faithfulness and the feasibility of counterfactual explanations in real-world applications.

Appendix A: Modeling Setups and Learned DAGs from ReX Causal Discovery

This appendix summarizes the experimental modeling setups and the ReX-based causal discovery procedure used in CausalCACTUS. All tabular datasets were pre-processed using the same standard procedures as in the main experiment. Causal graphs were learned per dataset using ReX (Renero et al., 2026) with default hyperparameters from the original implementation,⁴ which constructs a DAG $\mathcal{G}$ by training feature-wise regressors using bootstrapped SHAP attributions. We evaluated two levels of structural prior knowledge in ReX (two-tier setups: minimal and moderate priors for each dataset), producing two DAGs per dataset that were converted into adjacency matrices for downstream constraints in CausalCACTUS, attached as follows (Figs. 9, 10, 11, 12, 13). The detailed modeling configurations are available at our supporting website: <https://github.com/zhendong3wang/tabular-causal-counterfactuals>.

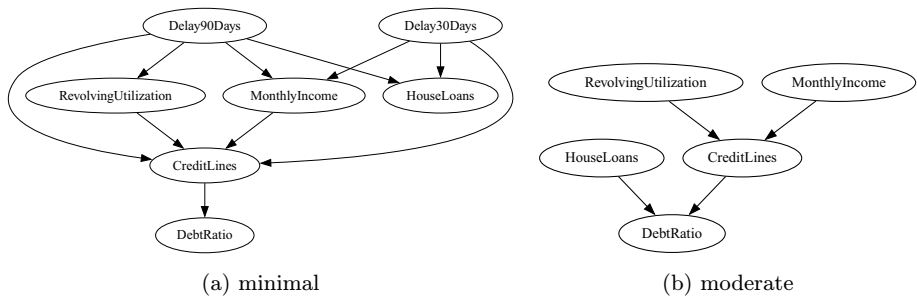


Fig. 9 Credit dataset DAGs (minimal vs. moderate priors)

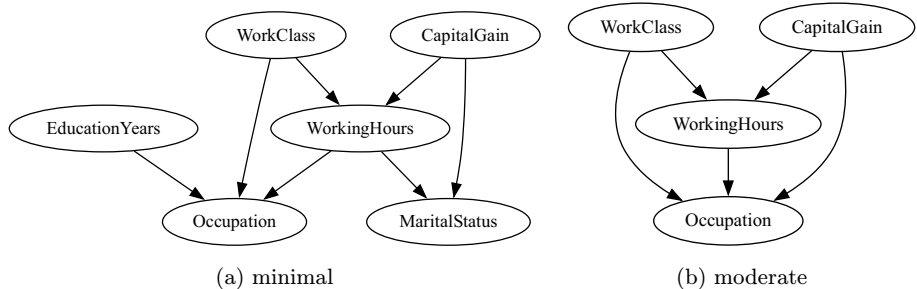


Fig. 10 Adult dataset DAGs (minimal vs. moderate priors)

⁴<https://github.com/renero/causalexplain/>.

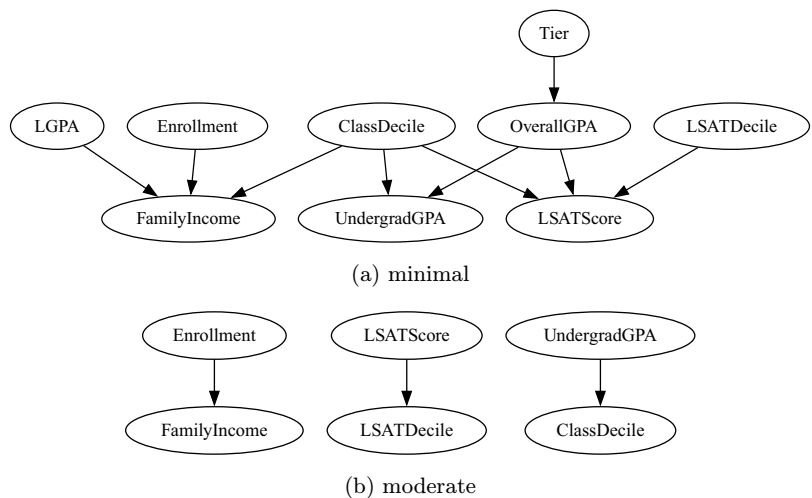


Fig. 11 Law dataset DAGs (minimal vs. moderate priors)

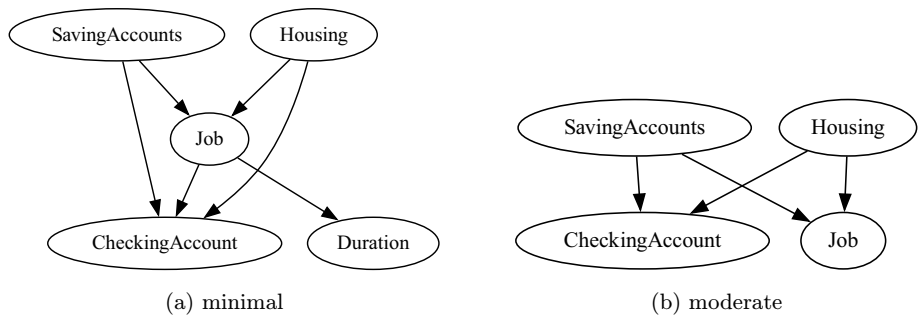


Fig. 12 German dataset DAGs (minimal vs. moderate priors)

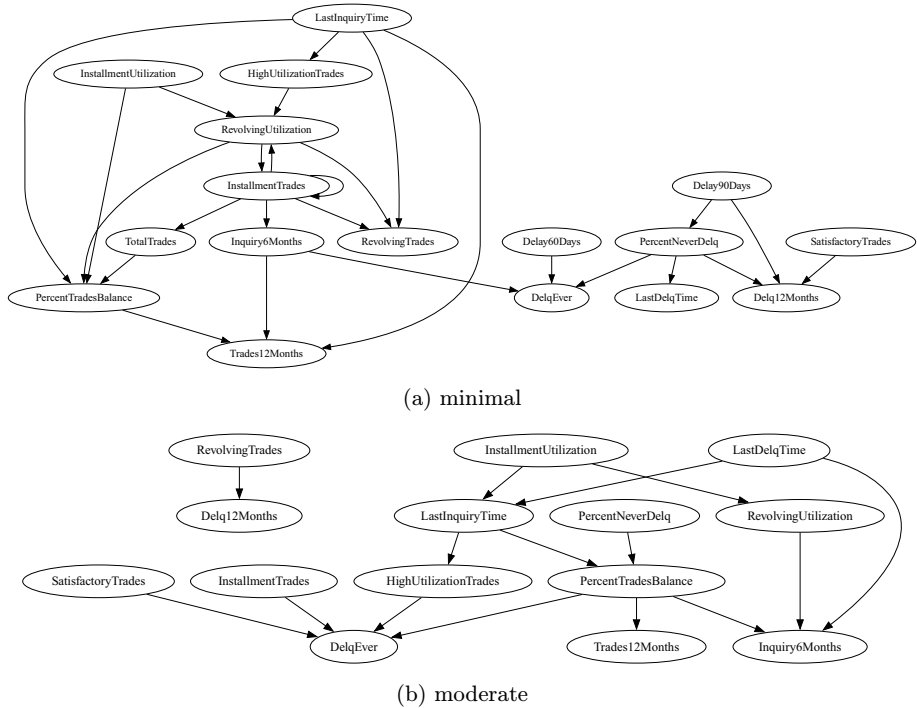


Fig. 13 HELOC dataset DAGs (minimal vs. moderate priors)

Appendix B: Extended Qualitative Results

This appendix extends the qualitative results presented in Sect. 6.4 to the four remaining (i.e., Adult, Law, German, and HELOC) datasets. Figure 14 visualizes the normalized mean absolute feature changes Δx using heatmaps, averaged over 50 randomly selected test instances and reported for each model and dataset. Consistent with the quantitative results, we observed that CausalCACTUS produced fewer amounts of feature changes (high compactness) and smaller feature modifications (low proximity), reflecting its improved performance relative to the baselines.

In addition, Figs. 15, 16, 17, and 18 show pairwise Spearman correlations $|\rho_s(\Delta x_i, \Delta x_j)|$ between feature changes across 50 randomly selected test samples for Adult, Law, German, and HELOC datasets, respectively. Black-outlined cells indicate parent–child pairs from the causal graph, where stronger dependencies are expected. We observed that CausalCACTUS mostly exhibited stronger correlations among causal dependency pairs while avoiding unnecessary correlations between causally unrelated pairs in the Adult, Law, and German datasets, indicating better alignment between the counterfactuals and the provided causal graphs.

Finally, Figs. 19, 20, 21, and 22 show representative counterfactual examples for Adult, Law, German, and HELOC datasets, respectively. In each figure, the first column reports the original feature values x_0 , while subsequent columns show the feature changes Δx proposed by each model. Across datasets, CausalCACTUS achieved the desired prediction with fewer

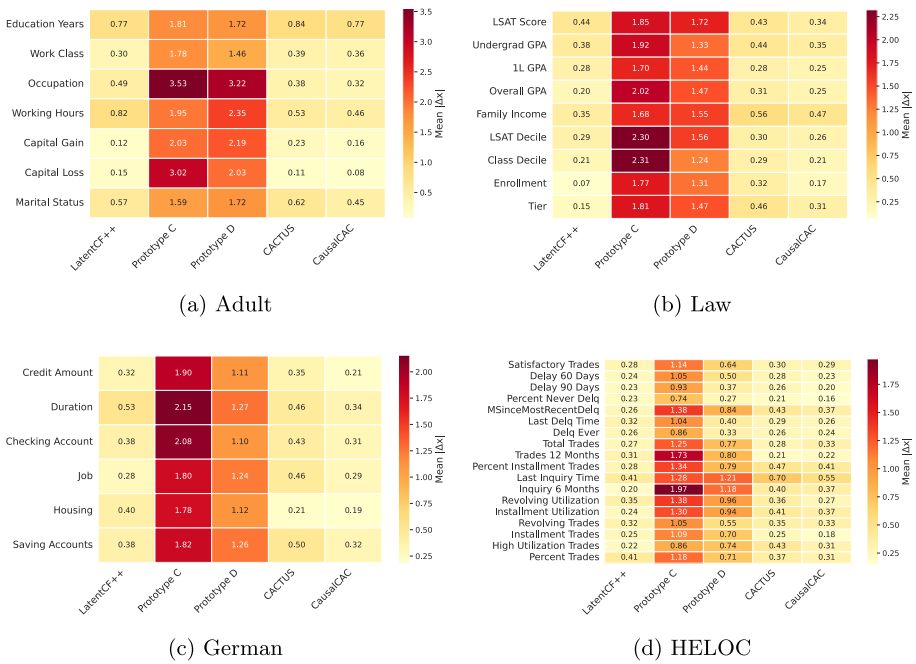


Fig. 14 Normalized mean absolute feature changes across CausalCACTUS and baseline models across all tabular datasets. Darker colors indicate larger average perturbations, highlighting differences in compactness and proximity among methods

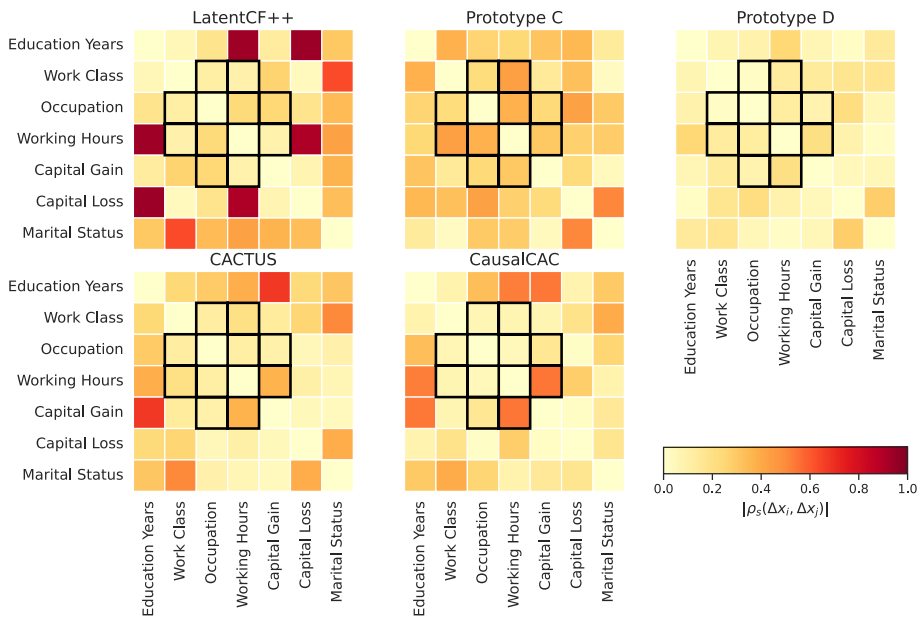


Fig. 15 Pairwise Spearman correlations of normalized feature changes across methods on the Adult dataset

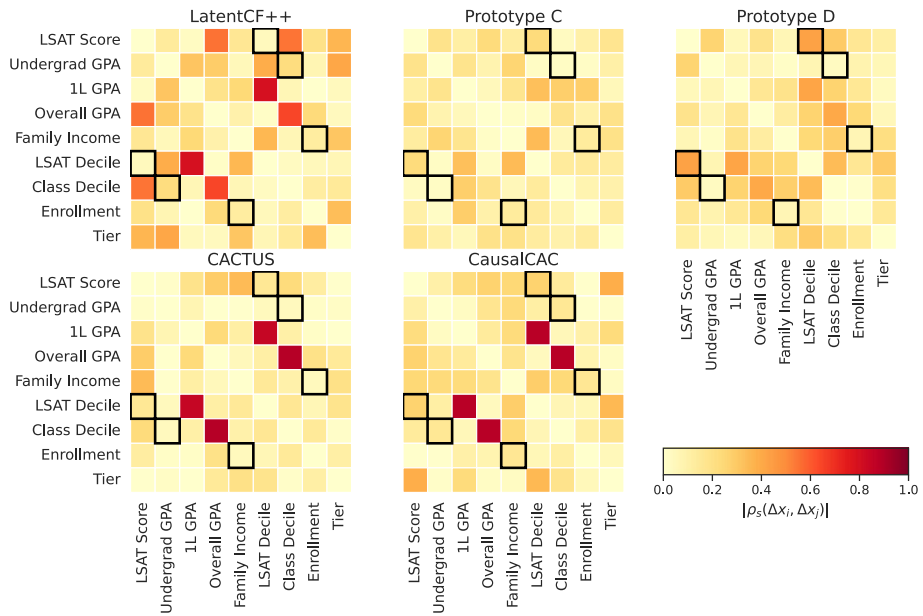


Fig. 16 Pairwise Spearman correlations of normalized feature changes across methods on the Law dataset

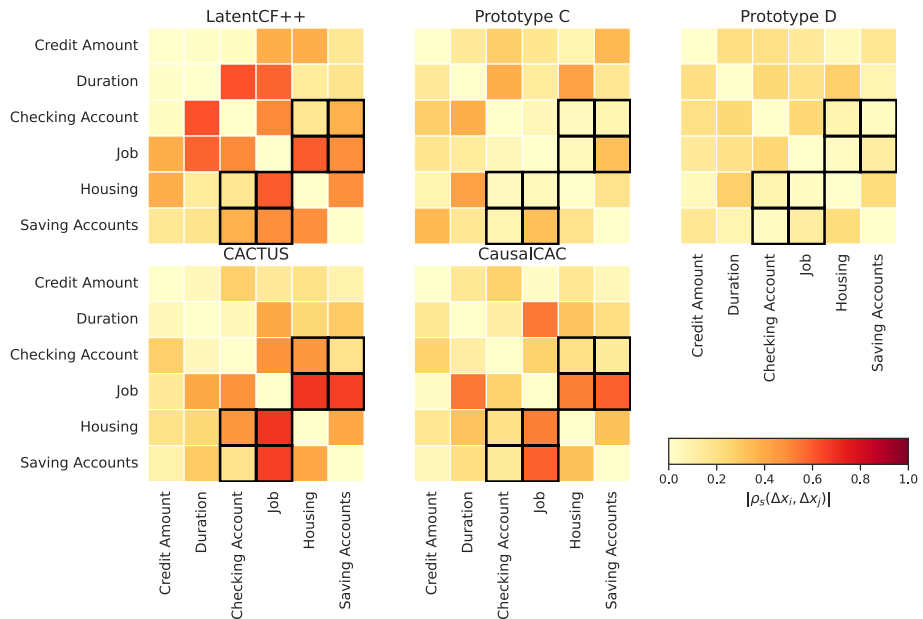


Fig. 17 Pairwise Spearman correlations of normalized feature changes across methods on the German dataset

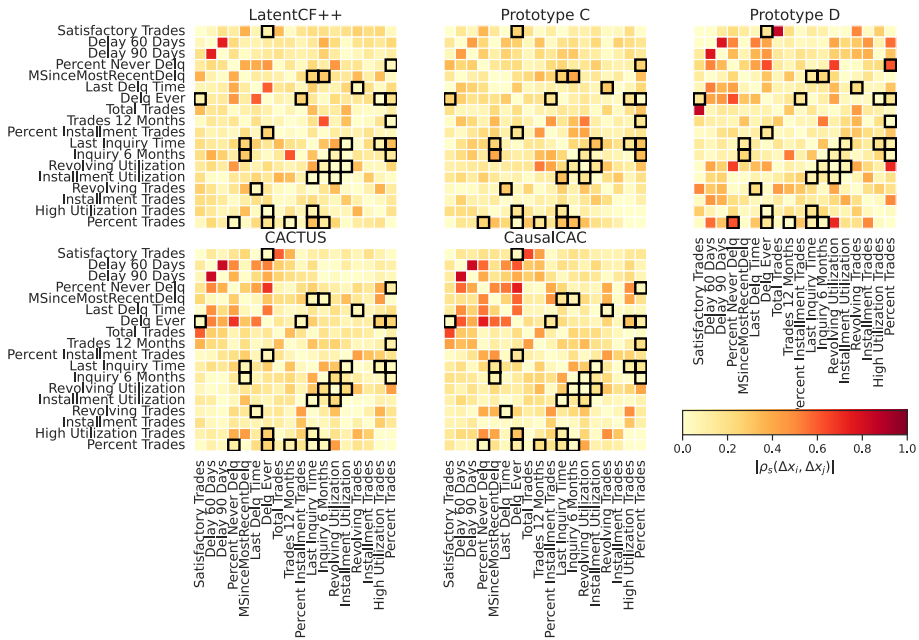


Fig. 18 Pairwise Spearman correlations of normalized feature changes across methods on the HELOC dataset

Education Years	11	+3	-7	-2	+4	+3
Marital Status	4	-1	+0	+0	-1	-1
Work Class	2	+0	+2	-4	-1	+0
Occupation	7	+1	-3	-3	+1	+1
Working Hours	35	+19	-39	-7	+4	+8
Capital Gain	0	-942	+16088	+43272	+1907	+1499
Capital Loss	0	+120	-1817	-790	+20	+16
	x0 (original)	Δ LatentCF++	Δ Prototype C	Δ Prototype D	Δ CACTUS	Δ CausalCAC

Fig. 19 Example of valid counterfactual explanations for the Adult dataset. The individual is female (the specified context), the causal graph is Fig. 10b, and the depicted changes alter the classifier's initial prediction from low-income ($\hat{y}_{la} = 0$) to high-income level ($\hat{y}_{cf,la} = 1$)

Family Income	3	+0	-0	+4	-1	-0
LSAT Score	15	+11	+36	+5	+7	+7
LSAT Decile	2	+5	-1	+10	+5	+5
Undergrad GPA	2	+1	+1	+0	+0	+0
1L GPA	-1	+1	-0	-0	+1	+1
Overall GPA	-0	+0	+2	+2	+0	+0
Class Decile	4	+2	-1	+2	+1	+1
Tier	0	+1	+1	+4	+1	+0
Enrollment	1	+0	+0	+1	+0	+0
	x0 (original)	Δ LatentCF++	Δ Prototype C	Δ Prototype D	Δ CACTUS	Δ CausalCAC

Fig. 20 Example of valid counterfactual explanations for the Law dataset. The individual is a non-white (the specified context), the causal graph is Fig. 11b, and the depicted changes alter the classifier's initial prediction from not passing ($\hat{y}_{la} = 0$) to passing the bar exam ($\hat{y}_{cf,la} = 1$)

Checking Account	1	+0	+3	-2	+0	+0
Saving Accounts	0	+1	-4	+1	+1	+1
Job	3	+0	+1	+0	-1	-1
Housing	0	+1	-4	+1	+0	+0
Credit Amount	5129	-2939	-9276	+6570	-3000	-1838
Duration	9	+3	-38	+0	+6	+12
	x0 (original)	Δ LatentCF++	Δ Prototype C	Δ Prototype D	Δ CACTUS	Δ CausalCAC

Fig. 21 Example of valid counterfactual explanations for the German dataset. The individual is a female (the specified context), the causal graph is Fig. 12b, and the depicted changes alter the classifier's initial prediction from high default risk ($\hat{y}_{la} = 1$) to low default risk ($\hat{y}_{cf,la} = 0$)

Total Trades -	20	+1	+3	+3	+1	+4
Percent Trades -	91	-13	+23	-21	+0	-1
Satisfactory Trades -	15	+2	-12	+8	+3	+6
Trades 12 Months -	1	+1	-4	+3	+0	+0
Revolving Trades -	6	-2	+4	+1	-4	-3
Installment Trades -	4	-2	+0	+1	+0	+0
Percent Installment Trades -	40	+1	+33	-4	+23	+23
Delq Ever -	2	+1	+0	+2	+1	+1
Delay 60 Days -	3	+1	-5	-1	+0	+0
Delay 90 Days -	3	+0	-5	-2	-1	-1
Percent Never Delq -	80	-11	-11	-1	-10	-10
MSinceMostRecentDelq -	2	-1	-15	+34	+11	+11
Last Delq Time -	0	+2	+11	+2	+3	+3
Last Inquiry Time -	0	+0	-6	+6	-0	-0
Inquiry 6 Months -	2	+0	+3	-4	+0	+0
Revolving Utilization -	35.00	+15.28	-7.07	+1.26	+2.87	+5.12
Installment Utilization -	82.00	+2.29	-16.12	+17.86	-0.56	-0.50
High Utilization Trades -	0.00	+1.00	+7.00	+2.00	-2.00	-1.00
	x0 (original)	Δ LatentCF++	Δ Prototype C	Δ Prototype D	Δ CACTUS	Δ CausalCAC

Fig. 22 Example of valid counterfactual explanations for the HELOC dataset. The context specifies that the individual's credit history is less than six years, and the causal graph is Fig. 13b. The depicted changes alter the classifier's initial prediction from high default risk ($\hat{y}_{la} = 1$) to low default risk ($\hat{y}_{cf,la} = 0$)

feature changes while better respecting the specified context and causal dependencies, consistent with the quantitative evaluation.

Author Contributions Z.W. conceived the study and led the development of the CausalCACTUS framework. Z.W., D.G., and S.C. designed and implemented the methodology. Z.W. and D.G. conducted the experiments, quantitative evaluation, and wrote the manuscript. J.E., S.C., and M.J. contributed to the experimental design and interpretation of results. All authors reviewed and approved the final manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Materials Availability Not Applicable.

Code Availability The source code is publicly available on GitHub: <https://github.com/zhendong3wang/tabular-causal-counterfactuals>.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical Approval and Consent to Participate Not applicable. All datasets used are publicly available and contain no sensitive information.

Consent for Publication All authors approve the publication of this work, including all content, data, and figures, in the Machine Learning Journal.

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Authors and Affiliations

Zhendong Wang¹ · Diego García² · José M. Enguita² · Saikat Chatterjee¹ · Magnus Jansson¹

✉ Zhendong Wang
zhwang@kth.se

Diego García
garciaperdiego@uniovi.es

José M. Enguita
jmenguita@uniovi.es

Saikat Chatterjee
sach@kth.se

Magnus Jansson
janssonm@kth.se

¹ KTH Royal Institute of Technology, Stockholm, Sweden

² University of Oviedo, Gijón, Spain